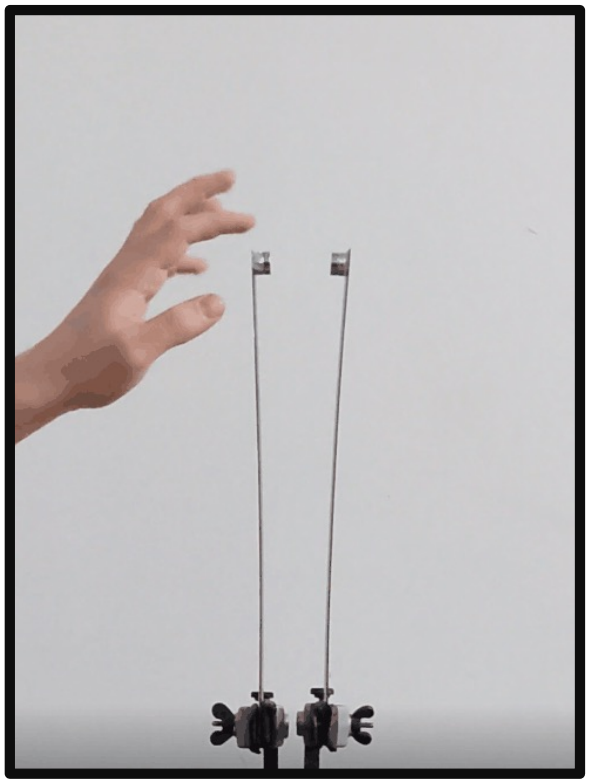
6. Magnetic-Mechanical Oscillator

Secure the lower ends of two identical leaf springs to a non-magnetic base and attach magnets to the upper ends such that they repel and are free to move. Investigate how the movement of the springs depends on relevant parameters.



Reporter: Dachi Tchotashvili

Phenomenon Demonstration



- *Magnets have to repel in order to be in stable equilibrium;*
- *During the motion, the Energy transfer from one magnet to another is happening;*
- *After some time, the system's energy damps.*

Phenomenon Demonstration

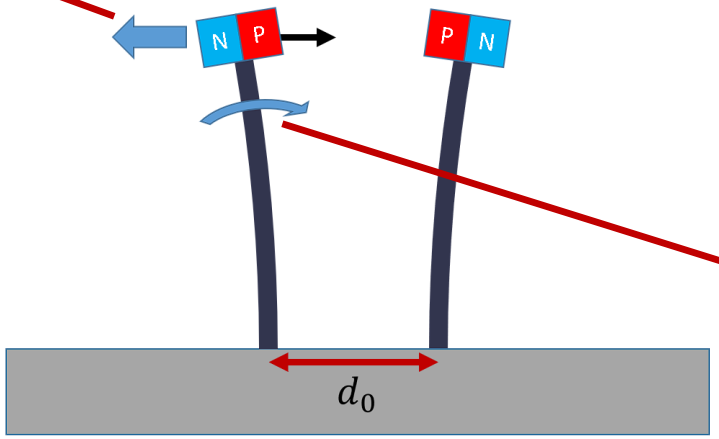


The most relevant parameters:

- Properties of magnets
- Dimensions and material of leaf spring
- Distance between leaf springs

Theoretical model

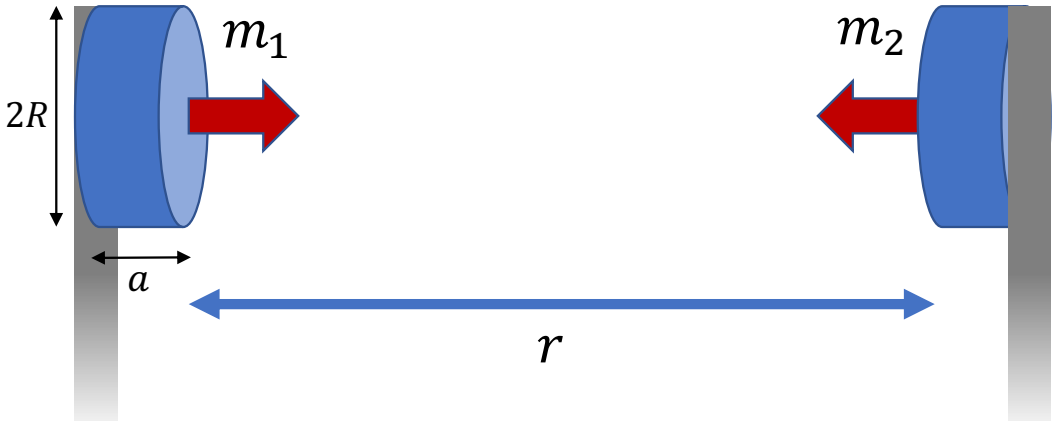
Magnetic force



Deformation of leaf spring

4

Calculating the Magnetic Force



$r \gg 2R, a$

$$F = \frac{3\mu_0 m_1 m_2}{2\pi r^4}$$

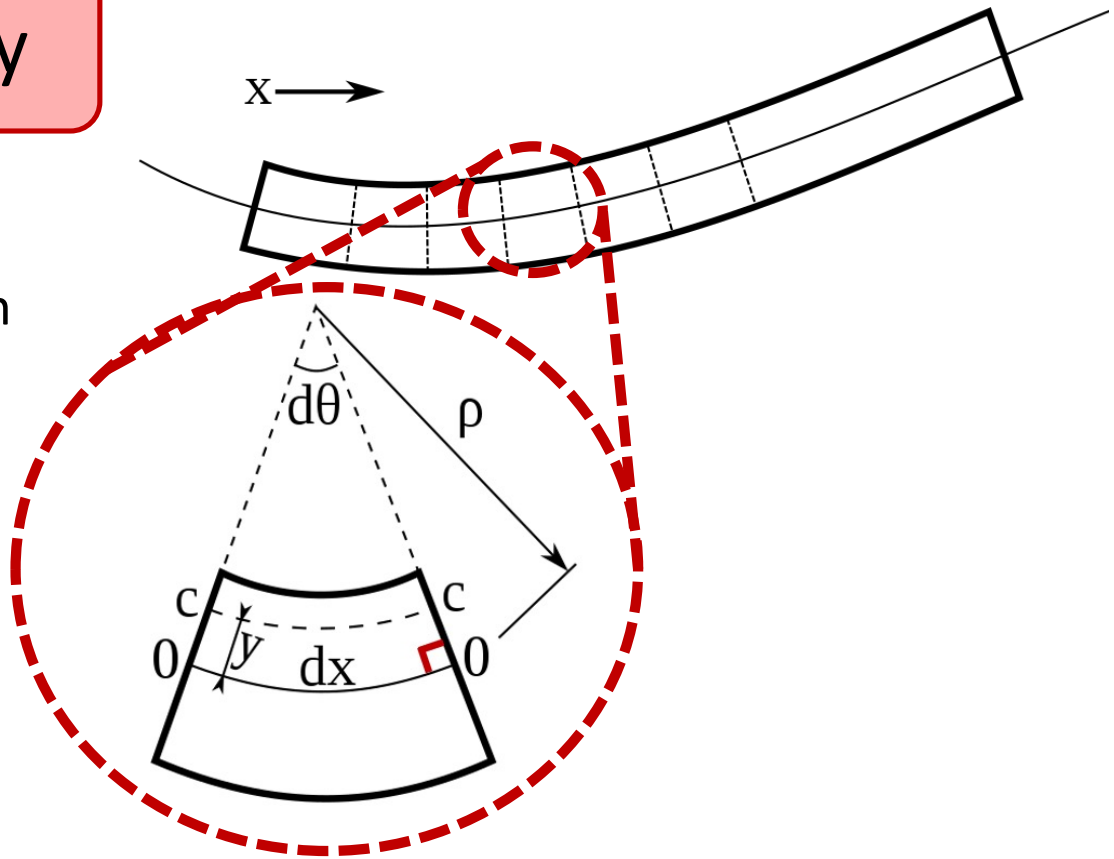
Distance between magnets

Product of magnetic moments

Deformation of leaf spring

https://en.wikipedia.org/wiki/Euler%E2%80%93Bernoulli_beam_theory#:~:text=Euler%E2%80%93Bernoulli%20beam%20theory%20(also,and%20deflection%20characteristics%20of%20beams.

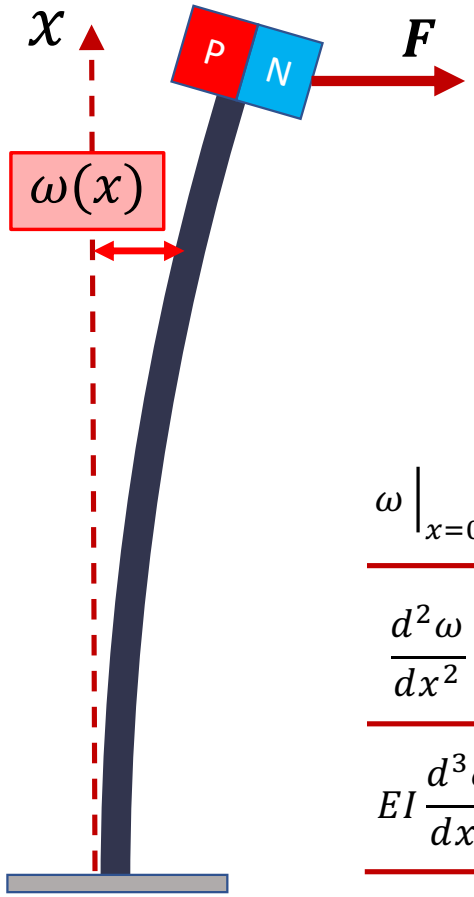
Euler-Bernoulli beam theory



! Approximations:

1. Neglected other types of deformation except bending
2. Bending is much less than the length of the beam
3. Width of the beam is much less than the length

Theoretical solution for the beam shape



$$EI\frac{d^4\omega}{dx^4} = 0$$

$E - Young's\ Modulos$
$I - second\ moment\ of\ inertia$
$\omega - deflection$
$x - coordinate$

Boundary conditions:

$\omega\big|_{x=0} = 0, \quad \frac{d\omega}{dx}\big|_{x=0} = 0$ — One end is fixed

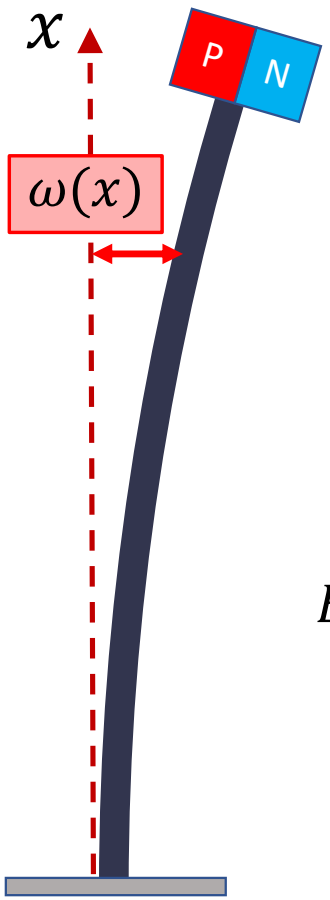
$\frac{d^2\omega}{dx^2}\big|_{x=L} = 0,$ — Free end has 0 torque applied

$EI\frac{d^3\omega}{dx^3}\big|_{x=L} = F$ — Force is acting on the free end

$$\omega(x) = \frac{F}{6EI}(3Lx^2 - x^3)$$

$$F = \frac{3EI}{L^3}\omega(L)$$

Measuring the effective mass



$$\omega(L) = \frac{F}{6EI} \cdot 2L^3 = \frac{FL^3}{3EI} \rightarrow F = \frac{3EI\omega(L)}{L^3} \rightarrow \omega = \frac{\omega(L)}{2L^3}(3Lx^2 - x^3)$$

$$v = \dot{\omega} = \frac{\dot{\omega}_L}{2L^3}(3Lx^2 - x^3)$$

Kinetic energy for some dx $dE_k = \frac{dmv^2}{2}$ *Whereas* $dm = \frac{dx}{L} m_{\text{ზოლი}}$

$$E_k = \int_0^L \frac{1}{2}\frac{dx}{L} m \frac{\dot{\omega}_L{}^2}{8L^6}(3Lx^2 - x^3)^2 = \frac{m\dot{\omega}_L{}^2}{8L^7}\left(9L^2\frac{x^5}{5} + \frac{x^7}{7} - Lx^6\right) = \frac{m\dot{\omega}_L{}^2}{2} \cdot \frac{33}{140}$$

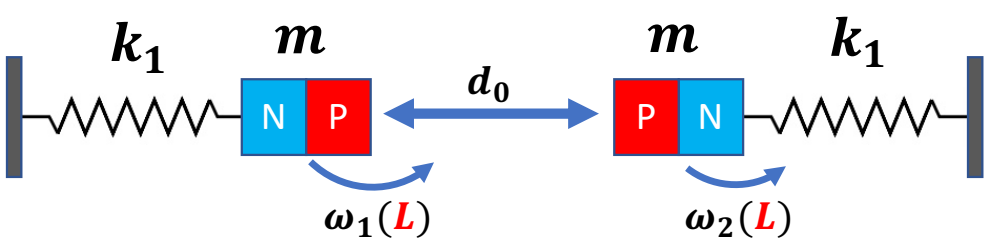
Effective mass of the spring $M = \frac{33}{140} m_{spring}$

Theoretical solution for the static beam

Calculating the k coefficient of leaf spring

$$F = \frac{3EI\omega(L)}{L^3}$$

$$k_1 = \frac{3EI}{L^3}$$



Magnetic force: $\frac{3\mu_0 m_1 m_2}{2\pi(\underbrace{d_0 + \omega_2(L) - \omega_1(L)}_{r})^4}$

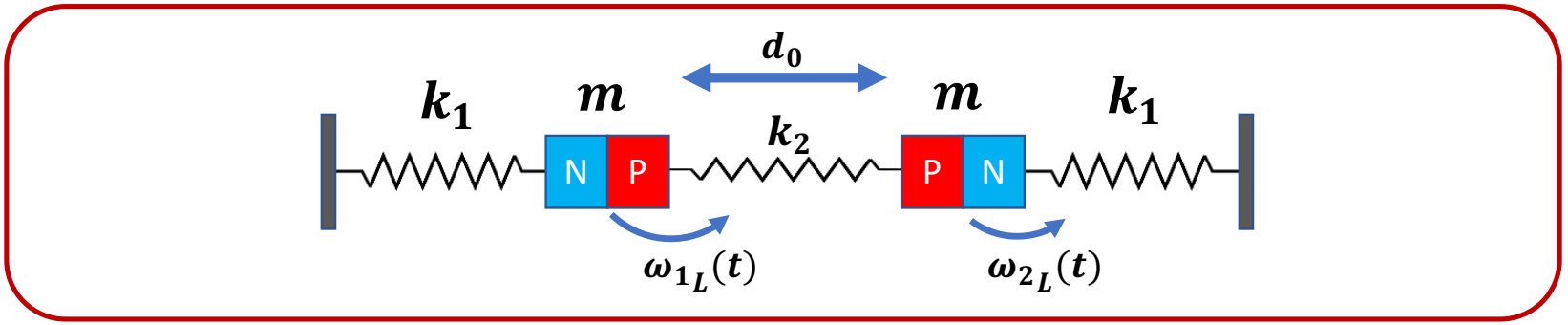
Linear approximation:

$d_0 \gg \omega_1(L), \omega_2(L)$

$$\frac{3\mu_0 m_1 m_2}{2\pi d_0^4} + \frac{6\mu_0 m_1 m_2}{\pi d_0^5}(\omega_1(L) - \omega_2(L))$$

C k_2

Coupled harmonic oscillation



Solution:

$$\omega_{1_L}(t) = e^{-\frac{\eta t}{2M}}\left[A_1 cos\left(\sqrt{\frac{k_1}{M}-\frac{\eta^2}{4M}}t+\varphi_1\right)+A_2\cos\left(\sqrt{\frac{k_1+2k_2}{M}-\frac{\eta^2}{4M}}t+\varphi_2\right)\right]+\underbrace{\frac{C}{k_1+2k_2}}_{x_0}$$

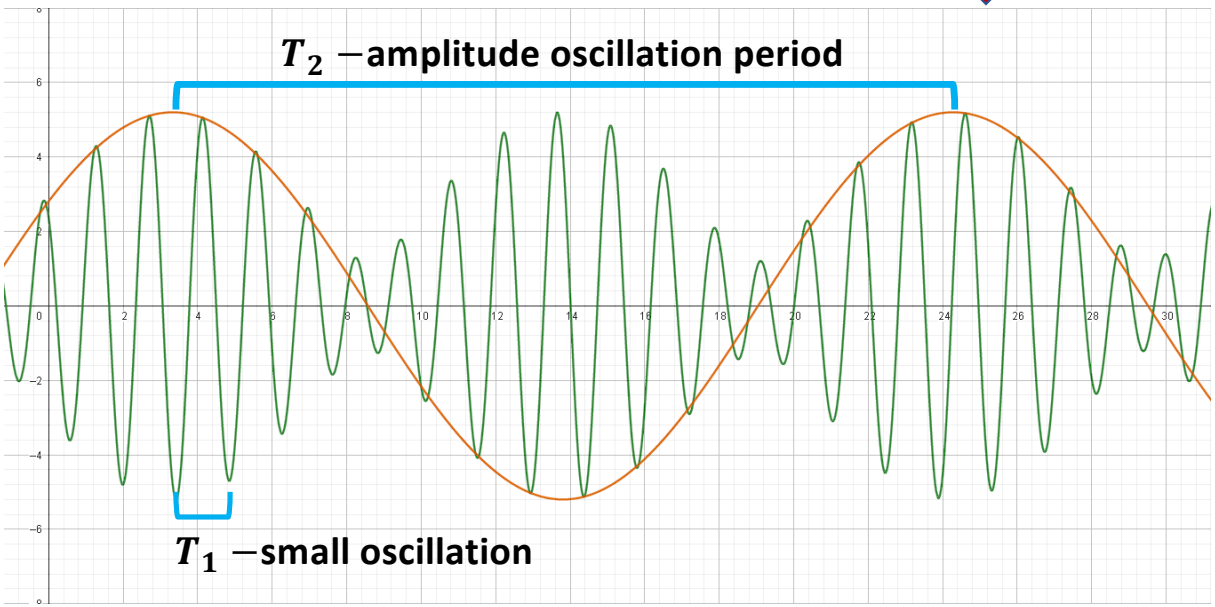
$$\omega_{2_L}(t) = e^{-\frac{\eta t}{2M}}\left[A_1 cos\left(\sqrt{\frac{k_1}{M}-\frac{\eta^2}{4M}}t+\varphi_1\right)-A_2\cos\left(\sqrt{\frac{k_1+2k_2}{M}-\frac{\eta^2}{4M}}t+\varphi_2\right)\right]-\frac{C}{k_1+2k_2}$$

Initial conditions:

$$\omega_{1_L}(0) = A$$
$$\dot{\omega}_{1_L}(0) = 0$$
$$\omega_{2_L}(0) = 0$$
$$\dot{\omega}_{2_L}(0) = 0$$

Theoretical solution for the dynamic beam

$$\omega_{1_L}(t) = e^{-\frac{\eta t}{2M}}\left[B_1 cos\left(\sqrt{\frac{k_1}{M}-\frac{\eta^2}{4M}}\,t+\varphi_1\right)+B_2\cos\left(\sqrt{\frac{k_1+2k_2}{M}-\frac{\eta^2}{4M}}\,t+\varphi_2\right)\right]$$



Frequencies investigated in data analysis

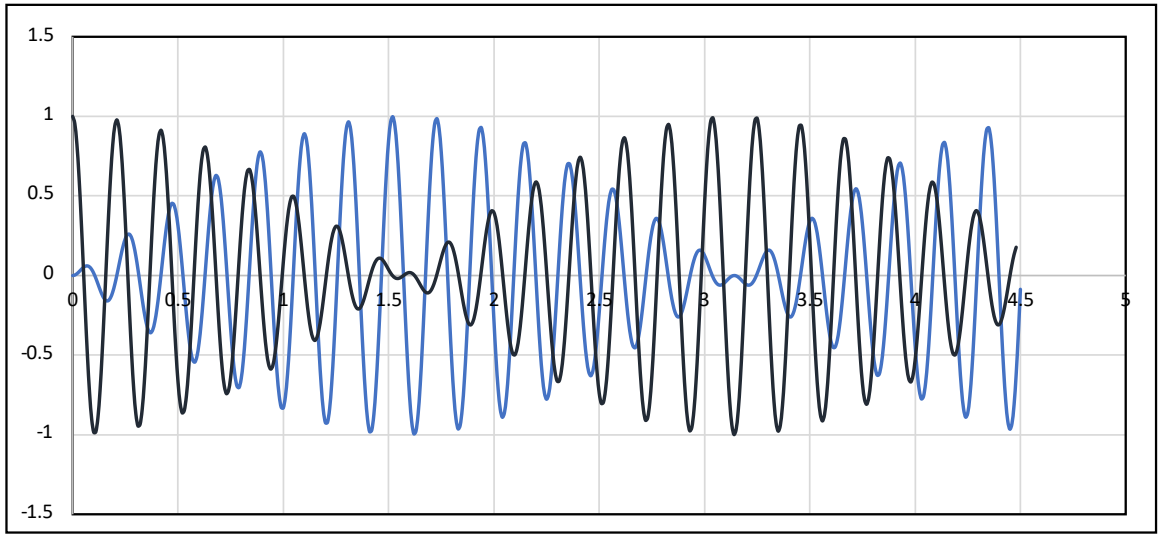
$$\Omega_1 = \frac{1}{2}\left(\sqrt{\frac{k_1+2k_2}{M}-\frac{\eta^2}{4M}}+\sqrt{\frac{k_1}{M}-\frac{\eta^2}{4M}}\right)$$

$$\Omega_1 = \frac{1}{2}\left(\sqrt{\frac{k_1+2k_2}{M}-\frac{\eta^2}{4M}}-\sqrt{\frac{k_1}{M}-\frac{\eta^2}{4M}}\right)$$

Coupled harmonic oscillation

$$x_1 = x_{m1}\cos(\omega_1 t + \varphi_1) + x_{m1}\cos(\omega_2 t + \varphi_2) \rightarrow x_1 = 2x_{m1}\cos\frac{\omega_1 + \omega_2}{2}\cos\frac{\omega_1 - \omega_2}{2}$$

$$x_2 = x_{m2}\cos(\omega_1 t + \varphi_1) - x_{m2}\cos(\omega_2 t + \varphi_2) \rightarrow x_2 = 2x_{m1}\sin\frac{\omega_1 + \omega_2}{2}\sin\frac{\omega_1 - \omega_2}{2}$$



Big period

$$T = \frac{2\pi}{\frac{\omega_2 - \omega_1}{2}} = \frac{4\pi}{\omega_2 - \omega_1}$$

small period

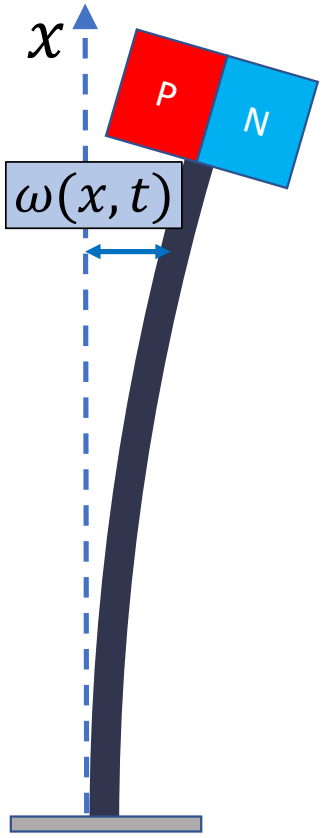
$$t = \frac{2\pi}{\frac{\omega_2 + \omega_1}{2}} = \frac{4\pi}{\omega_2 + \omega_1}$$

Final theoretical result

Displacement of the beam as a function of coordinate and time

$$\omega_1(x,t) = \left[e^{-\frac{\eta t}{2M}}\left(A_1 cos\left(\frac{\Omega_1 - \Omega_2}{2}t + \varphi_1\right) + A_2 \cos\left(\frac{\Omega_1 + \Omega_2}{2}t + \varphi_2\right)\right) + x_0\right]$$

$$\omega_2(x,t) = \left[e^{-\frac{\eta t}{2M}}\left(A_1 cos\left(\frac{\Omega_1 - \Omega_2}{2}t + \varphi_1\right) - A_2 \cos\left(\frac{\Omega_1 + \Omega_2}{2}t + \varphi_2\right)\right) - x_0\right]$$

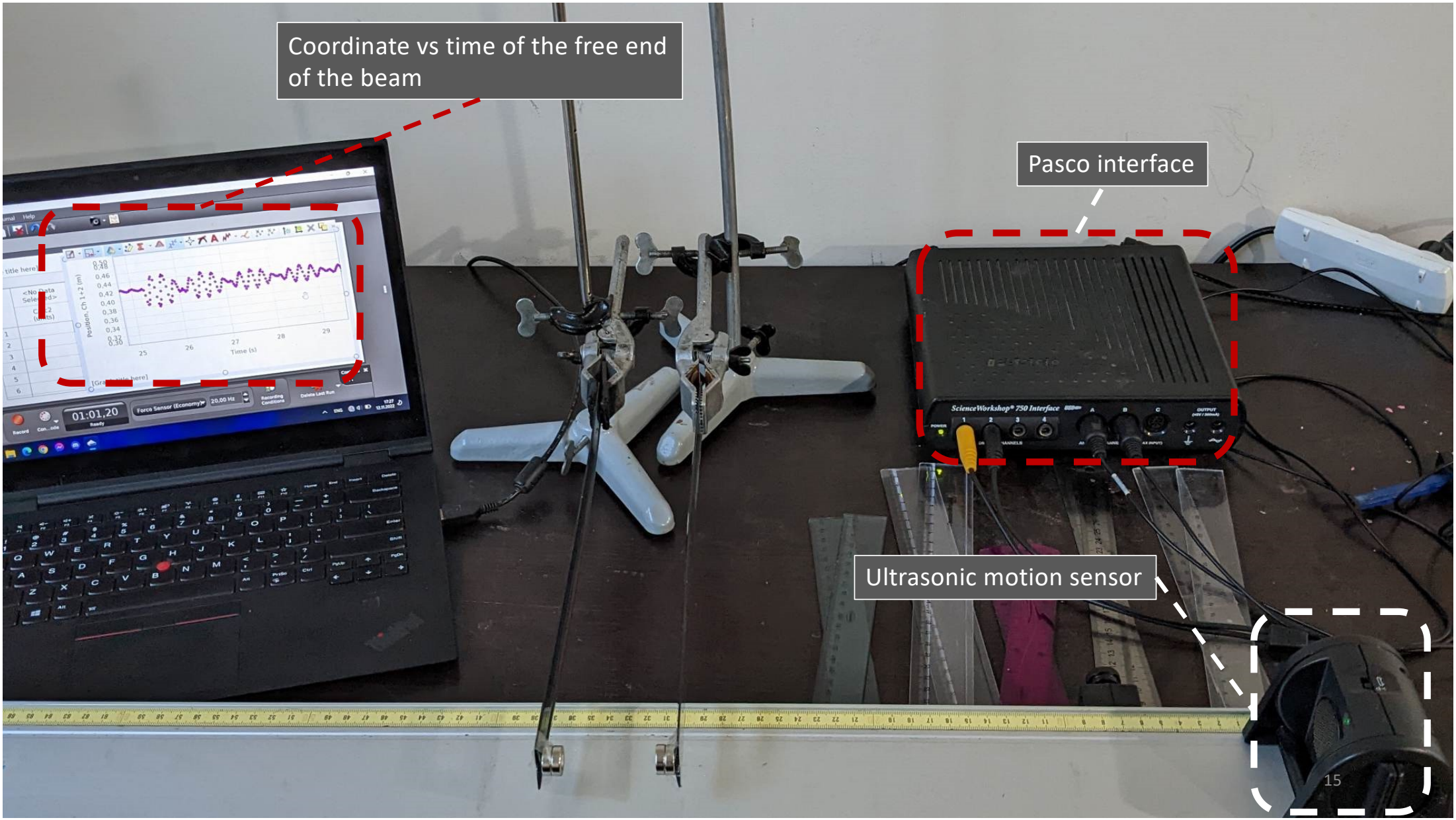


Characteristic frequencies

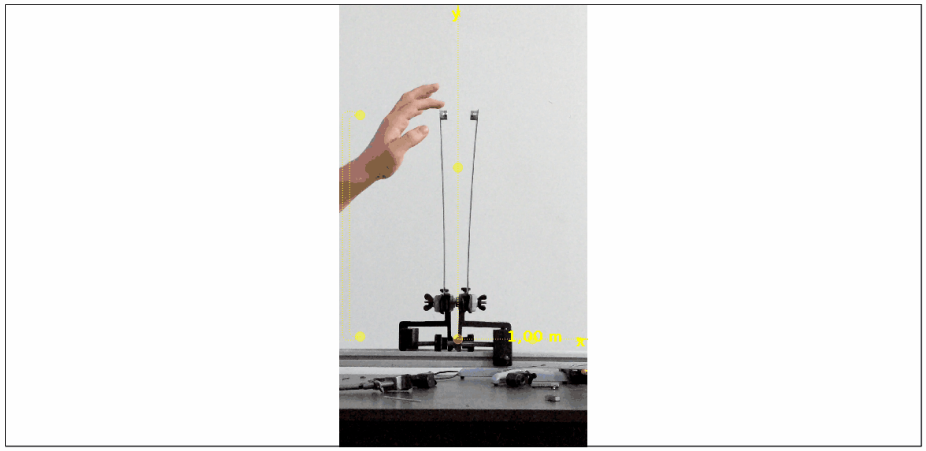
$$\boldsymbol{\Omega_1} = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} + \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

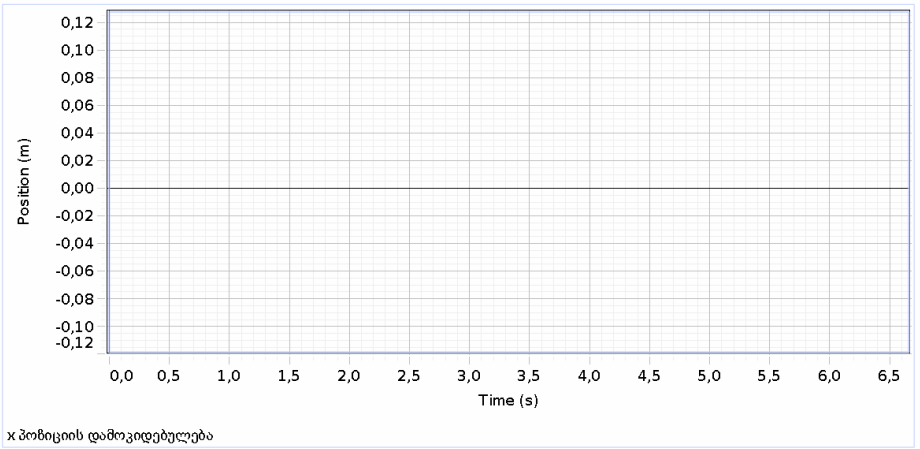
$$\boldsymbol{\Omega_2} = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} - \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

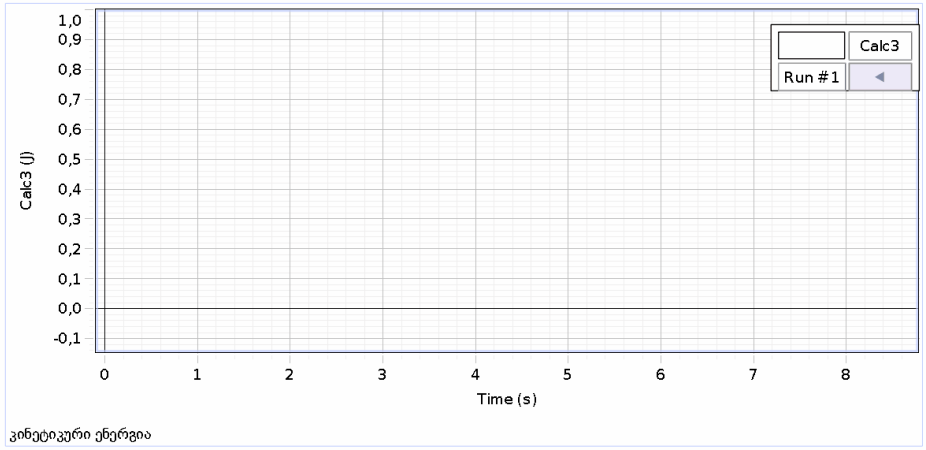
Experiments

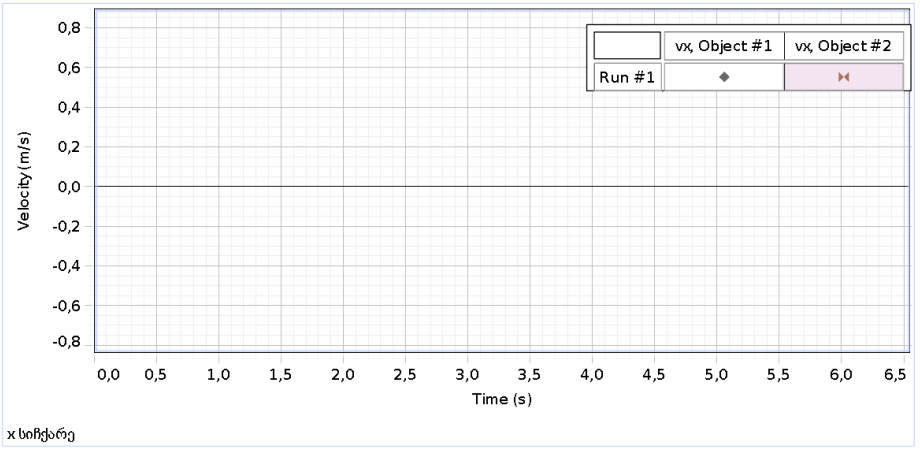

Coordinate vs time of the free end of the beam
Pasco interface
Ultrasonic motion sensor
ScienceWorkshop® 750 Interface
01:01,20
Time (s)

Second method of data taking

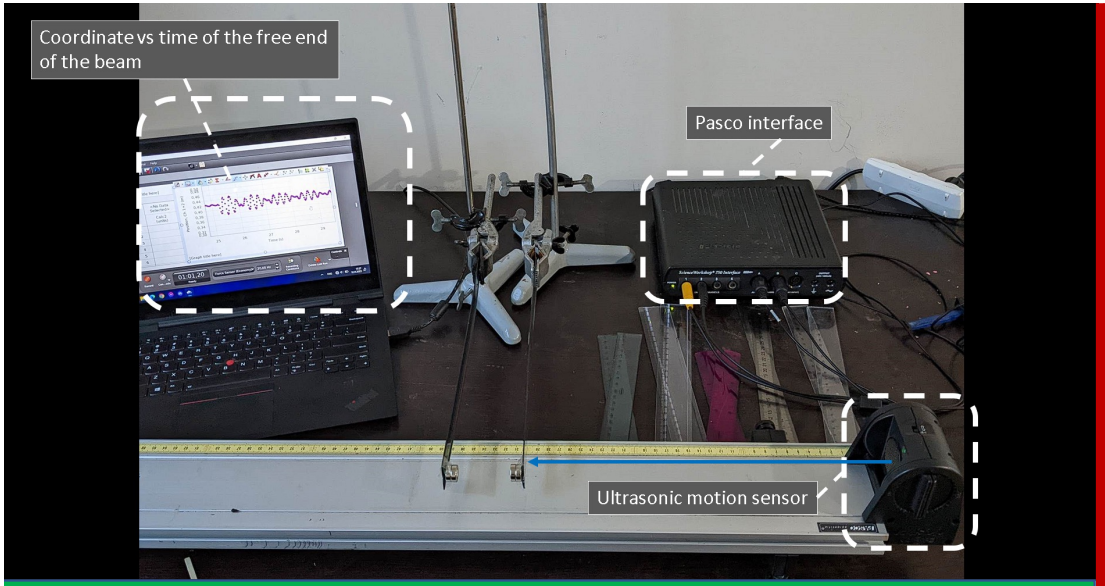


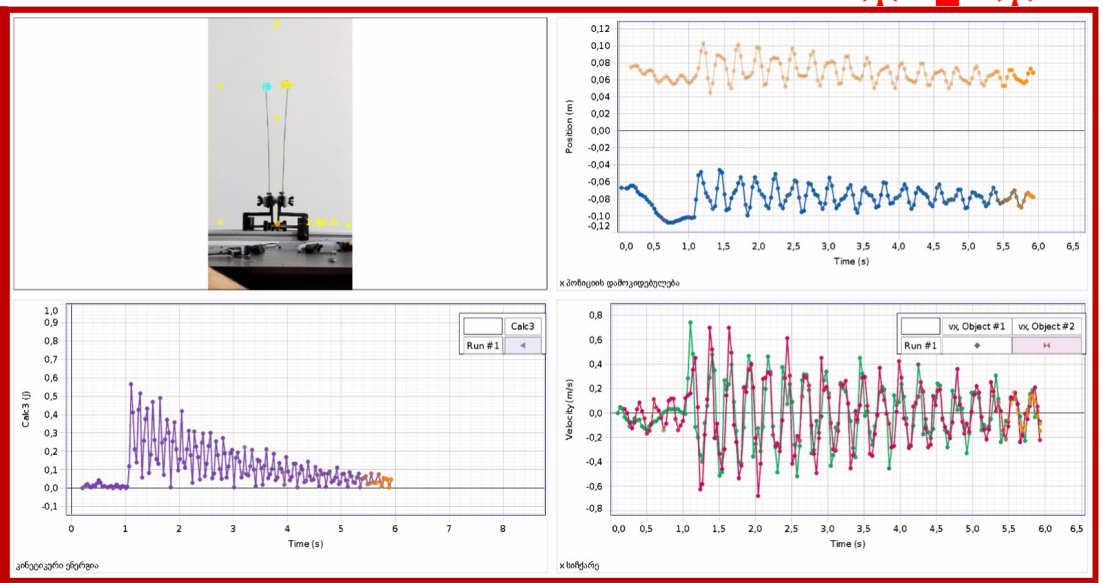






Comparison of 2 methods





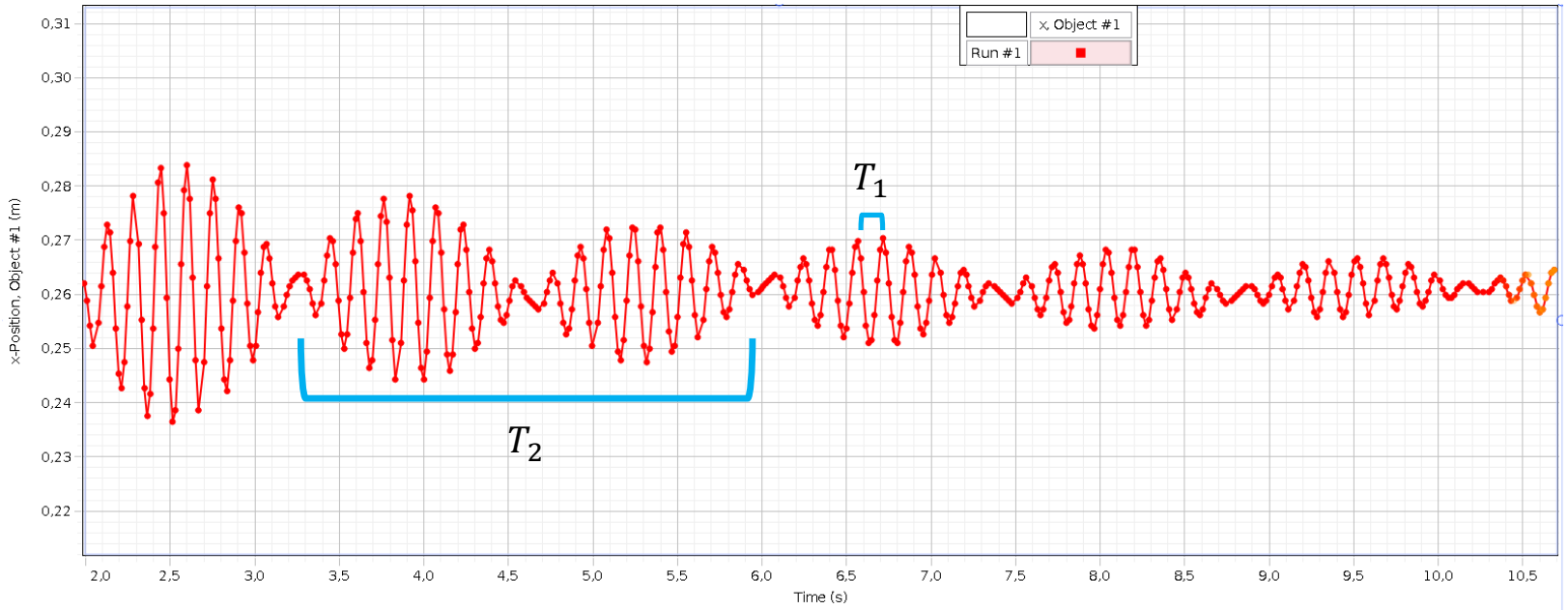
Faster to process, accurate

Can measure only one point of the beam

Optical and tracking error,

Can track whole length of the beam

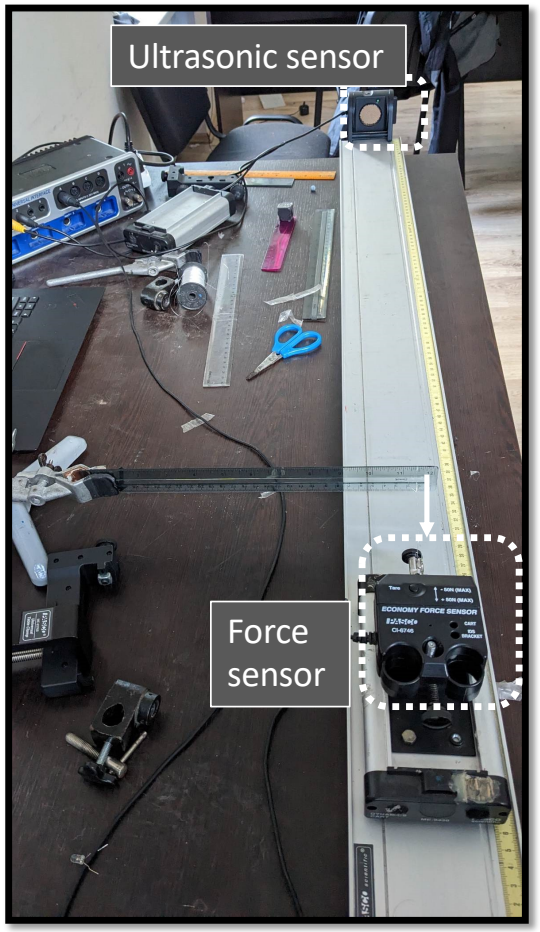
Measuring frequencies experimentally

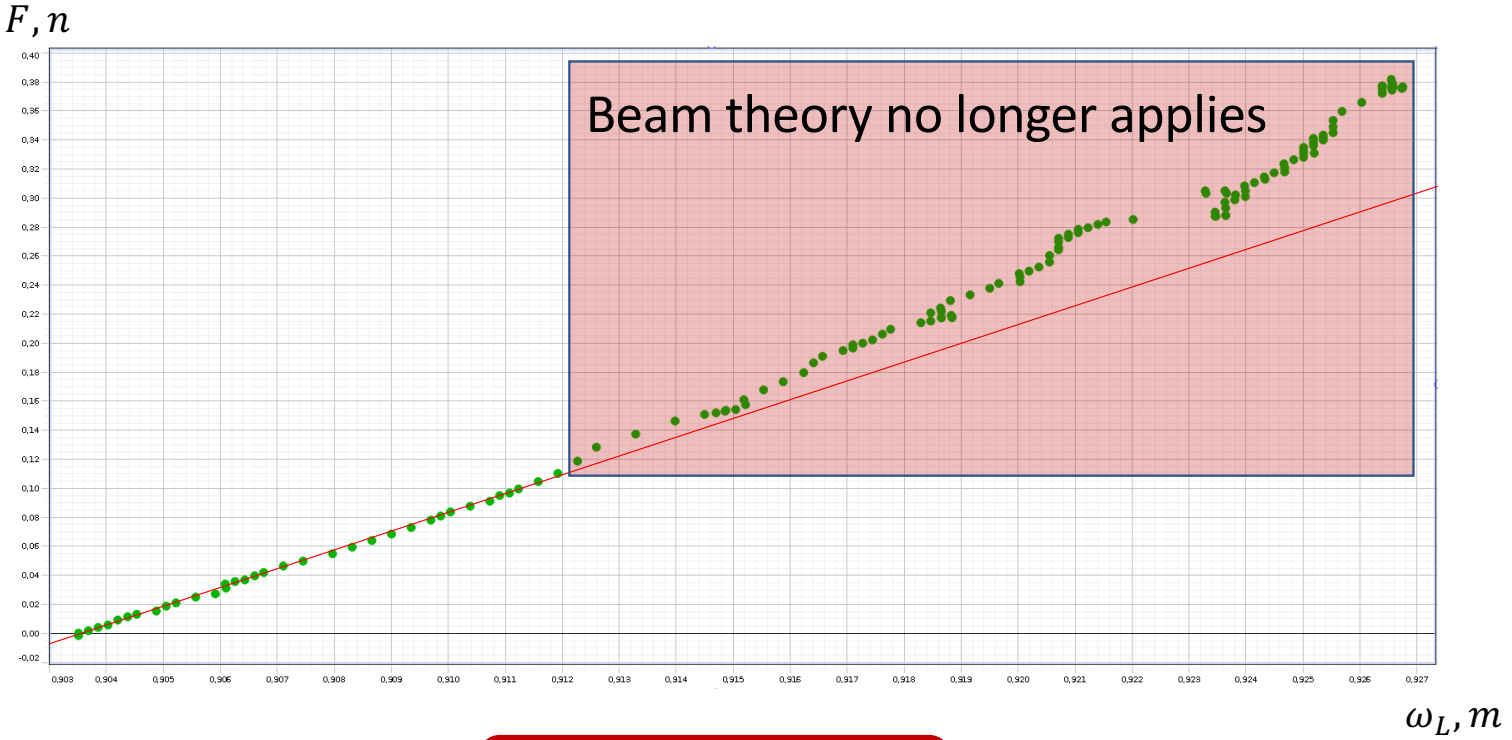


$$\Omega_1 = \frac{2\pi}{T_1}$$

$$\Omega_2 = \frac{2\pi}{T_2}$$

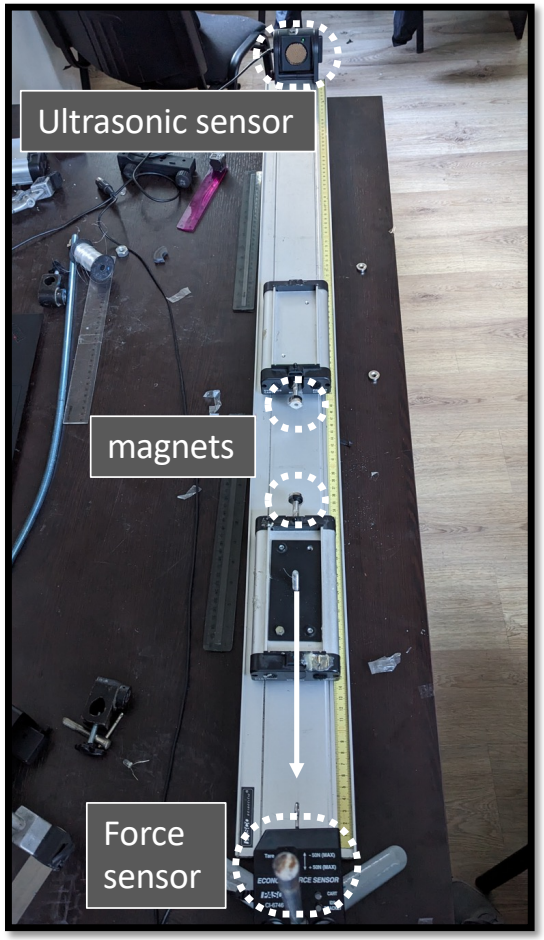
Experimental measurement of K1

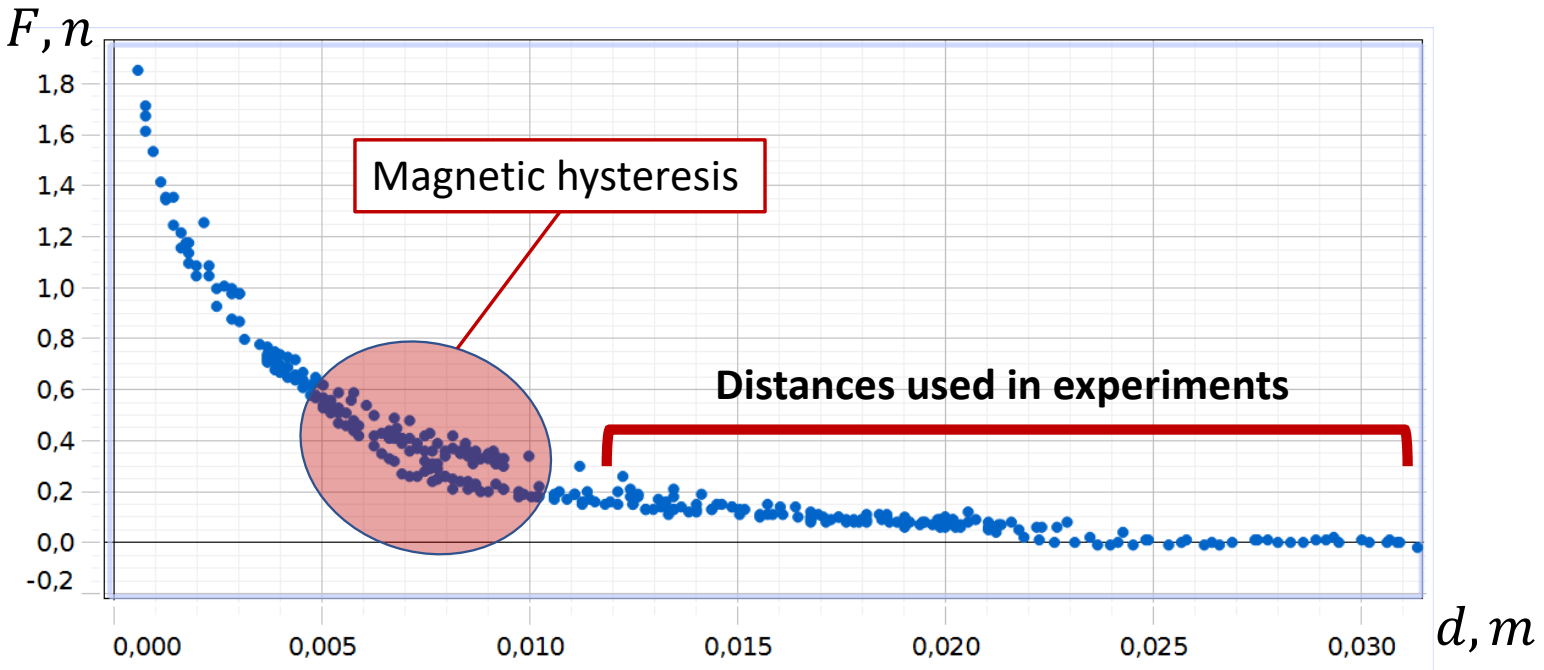




$$F = \frac{3EI}{L^3}\omega(L)$$

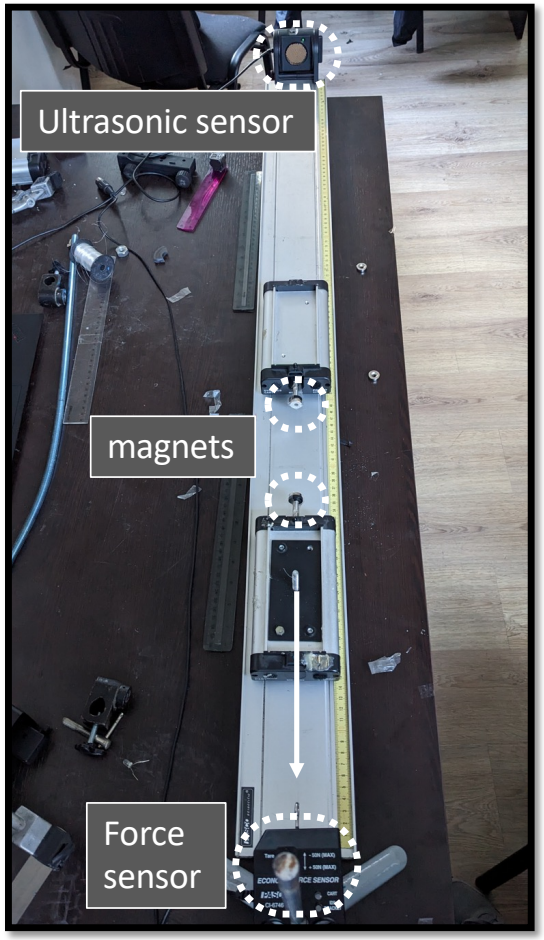
Experimental measurement of magnetic moment



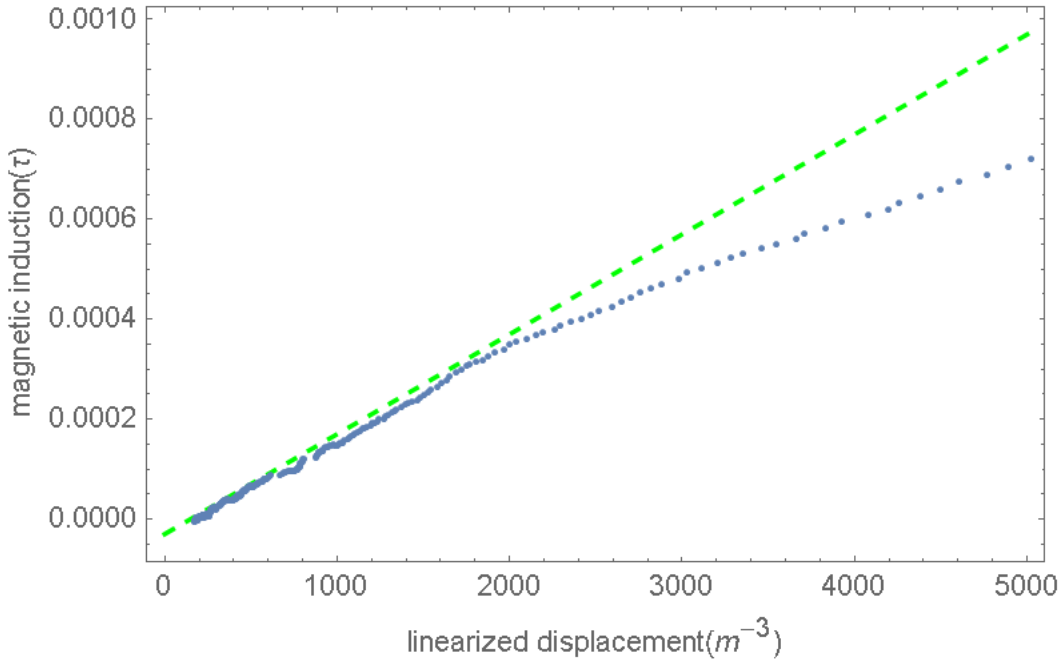


$$F = -\frac{3\mu_0 m_1 m_2}{2\pi d^4}$$

Experimental measurement of magnetic moment



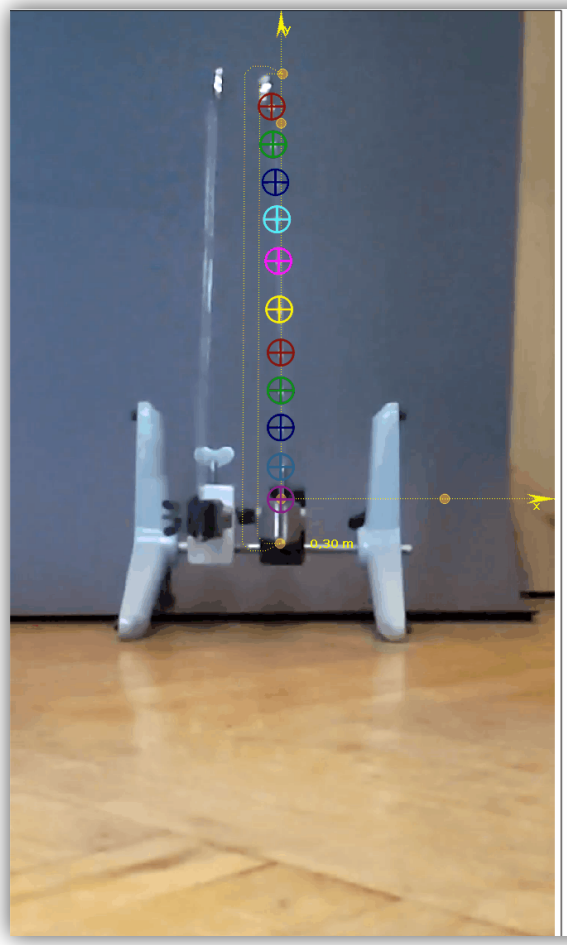
F, n

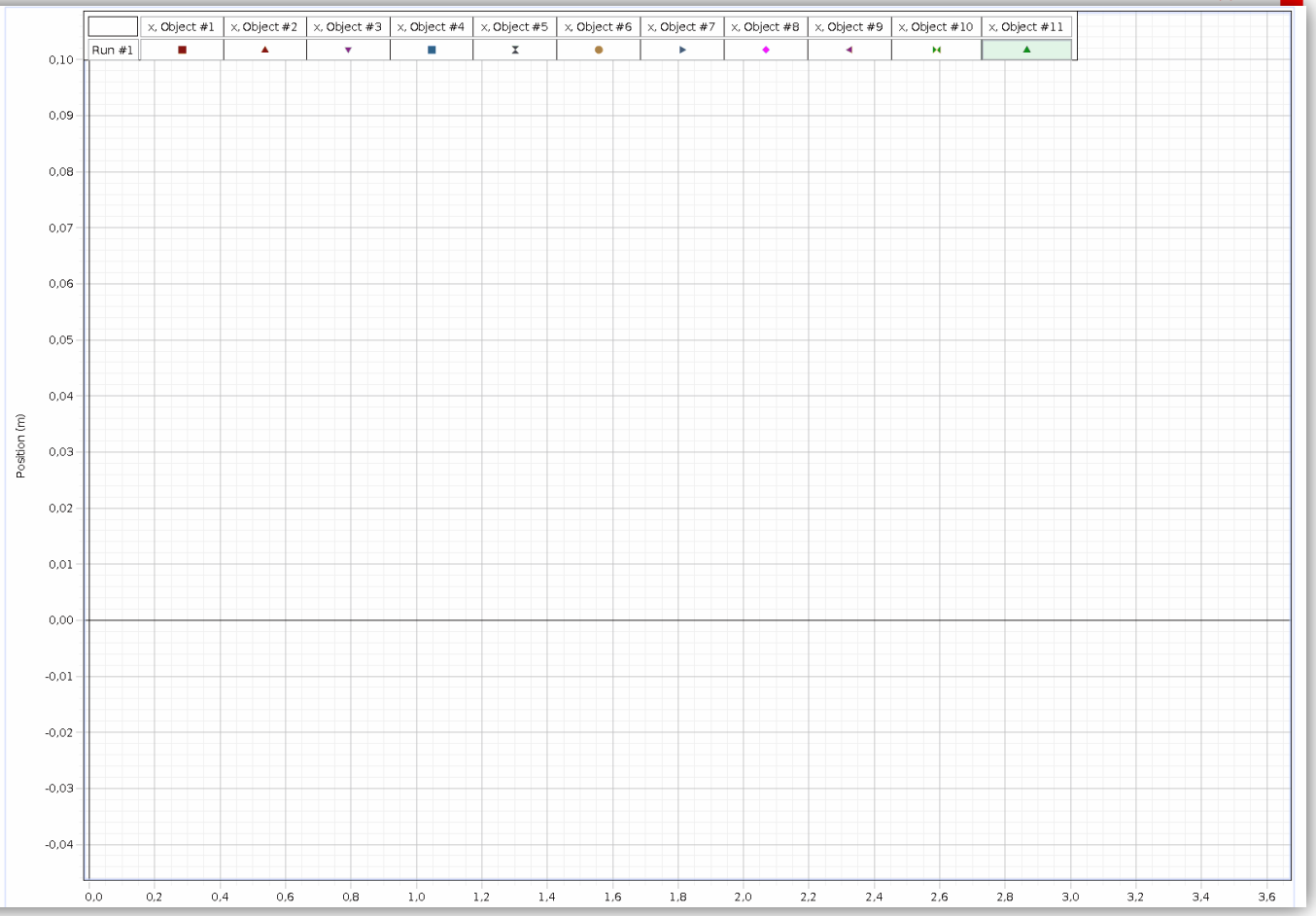


d, m

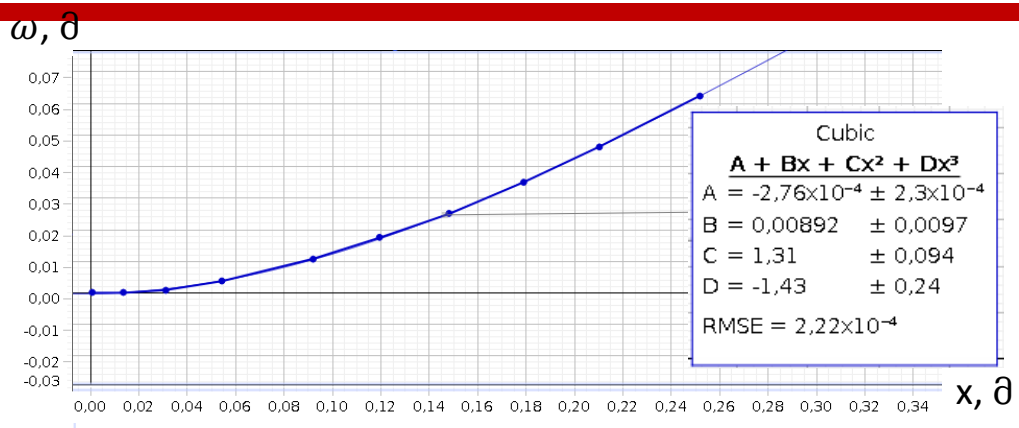
$$F = -\frac{3\mu_0 m_1 m_2}{2\pi d^4}$$

Investigation of the shape of the beam





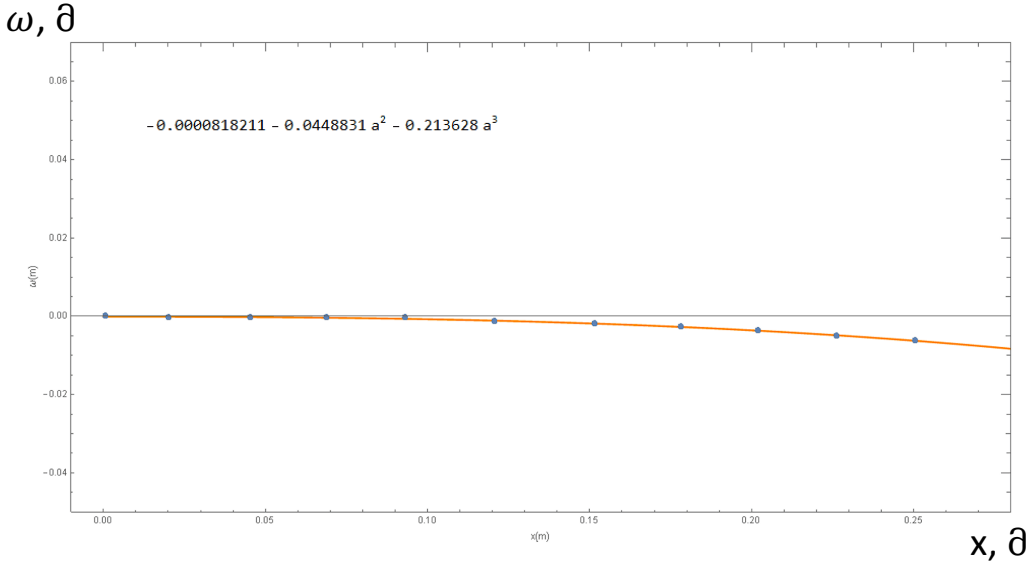
Investigation of the shape of the beam



Static beam shape

$C \approx -3LD$

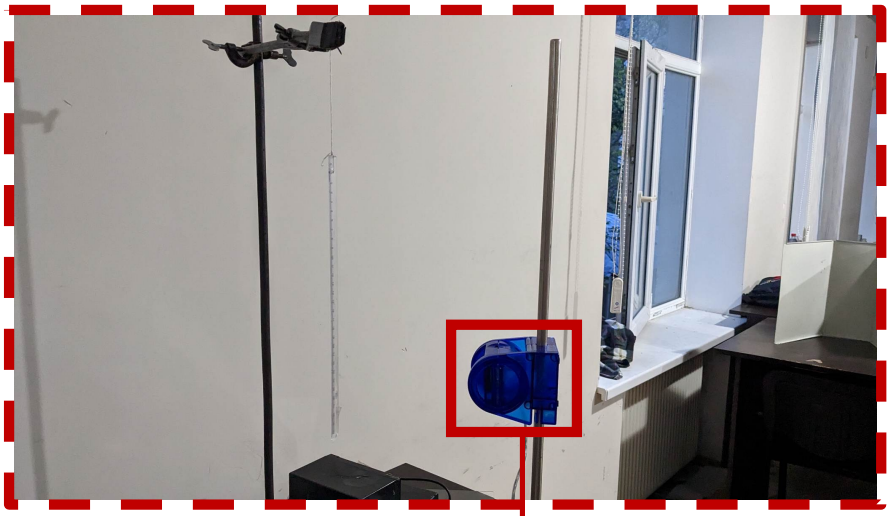
$$\omega = A(x^3 - 3Lx^2)$$



Dynamic beam shape

$C \approx -3LD$

Observing air friction coefficient



Pasco motion sensor

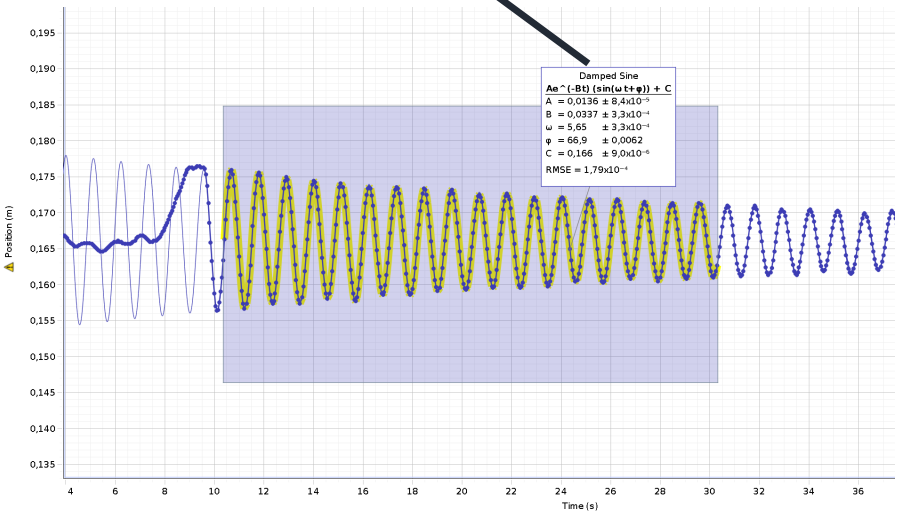
Damped Sine

$Ae^{(-Bt)}(\sin(\omega t+\varphi)) + C$

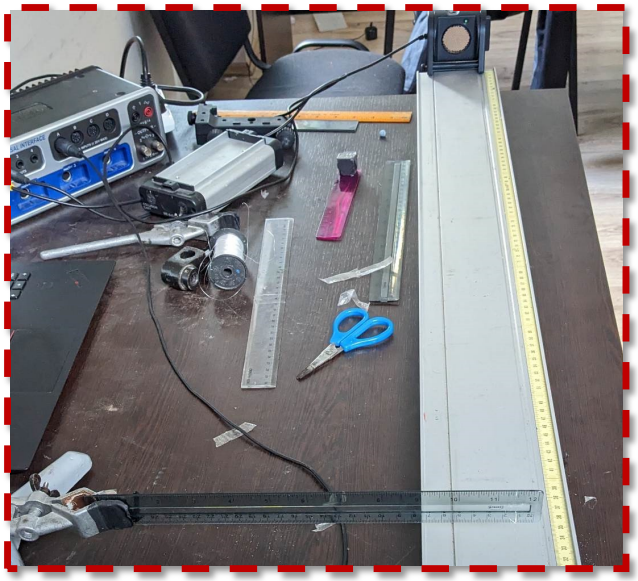
A	= 0,0136	± 8,4x10^{-5}
B	= 0,0337	± 3,3x10^{-4}
ω	= 5,65	± 3,3x10^{-4}
φ	= 66,9	± 0,0062
C	= 0,166	± 9,0x10^{-6}

RMSE = 1,79x10^{-4}

$$\beta = 0.0337$$



Observing bending friction coefficient



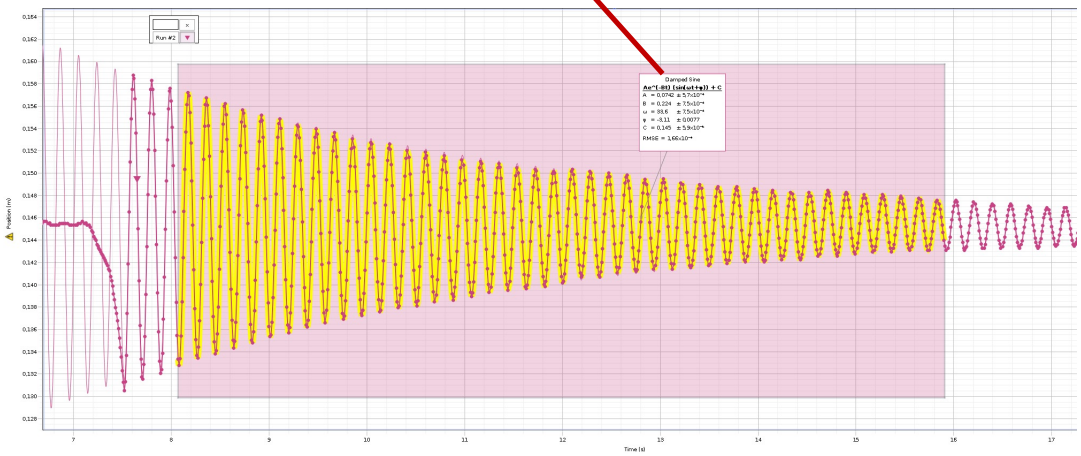
Damped Sine

$Ae^{(-Bt)}(\sin(\omega t+\varphi)) + C$

A	= 0,0742	± 5,7x10⁻⁴
B	= 0,224	± 7,5x10⁻⁴
ω	= 33,6	± 7,5x10⁻⁴
φ	= -3,11	± 0,0077
C	= 0,145	± 5,9x10⁻⁵

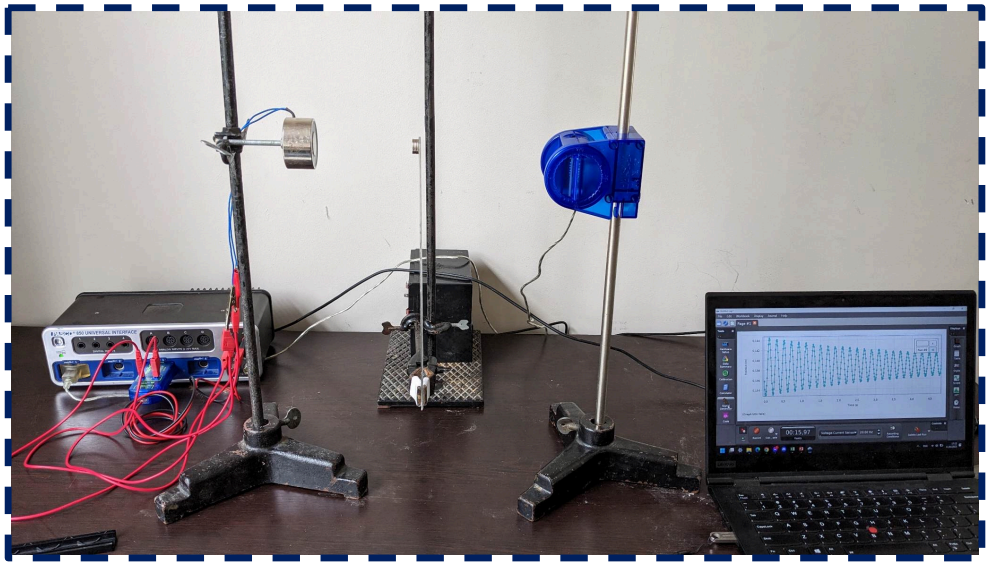
RMSE = 1,66x10⁻⁴

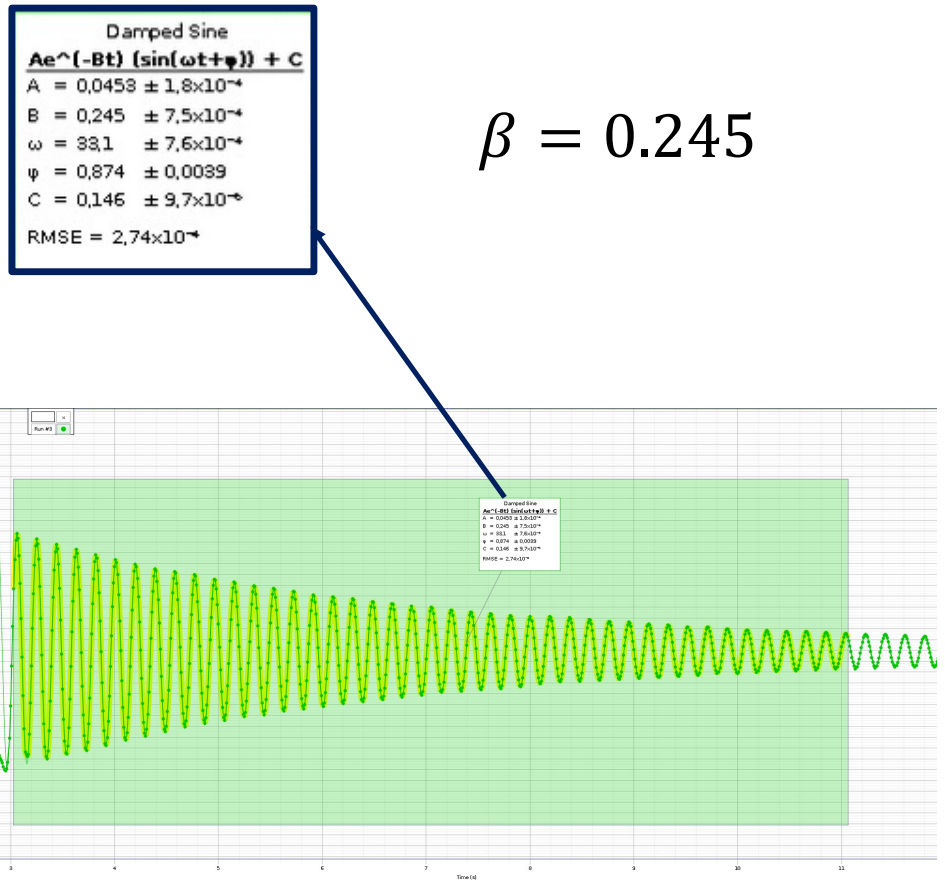
$$\beta = 0.224$$



but actual bending friction equals $0.224 - 0.0337 = 0.1903$

Observing magnetic friction coefficient





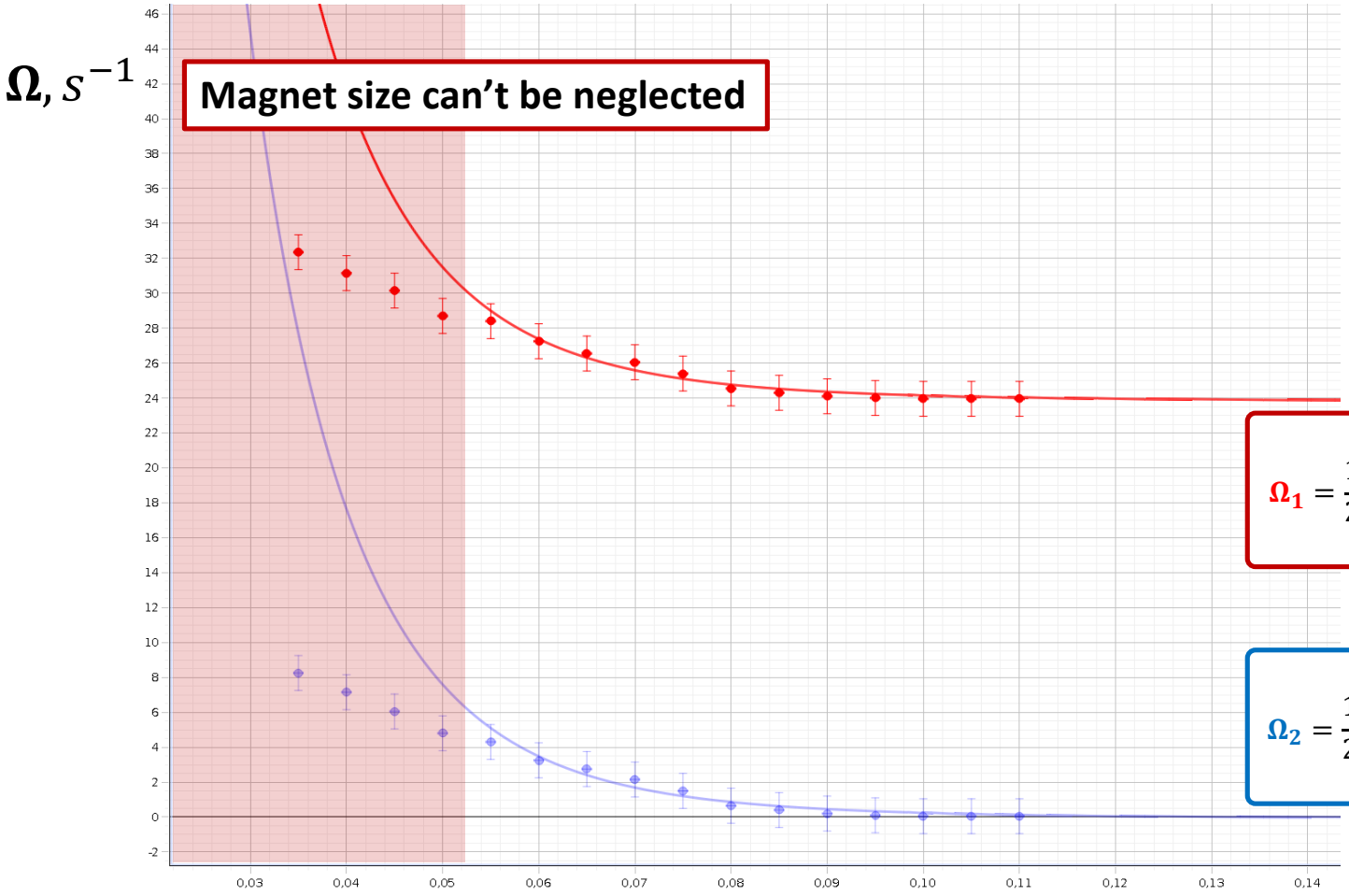
$\beta = 0.245$

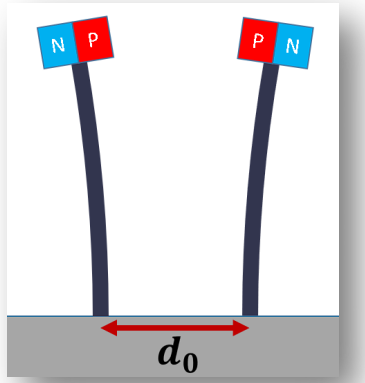
$magnetic\ friction\ coefficient\ equals - 0.245 - 0.224 = 0.021$

phenomenon | Theoretical model | experiment | conclusion

Data analysis

Ω_1, Ω_2 vs Distance between leaf springs

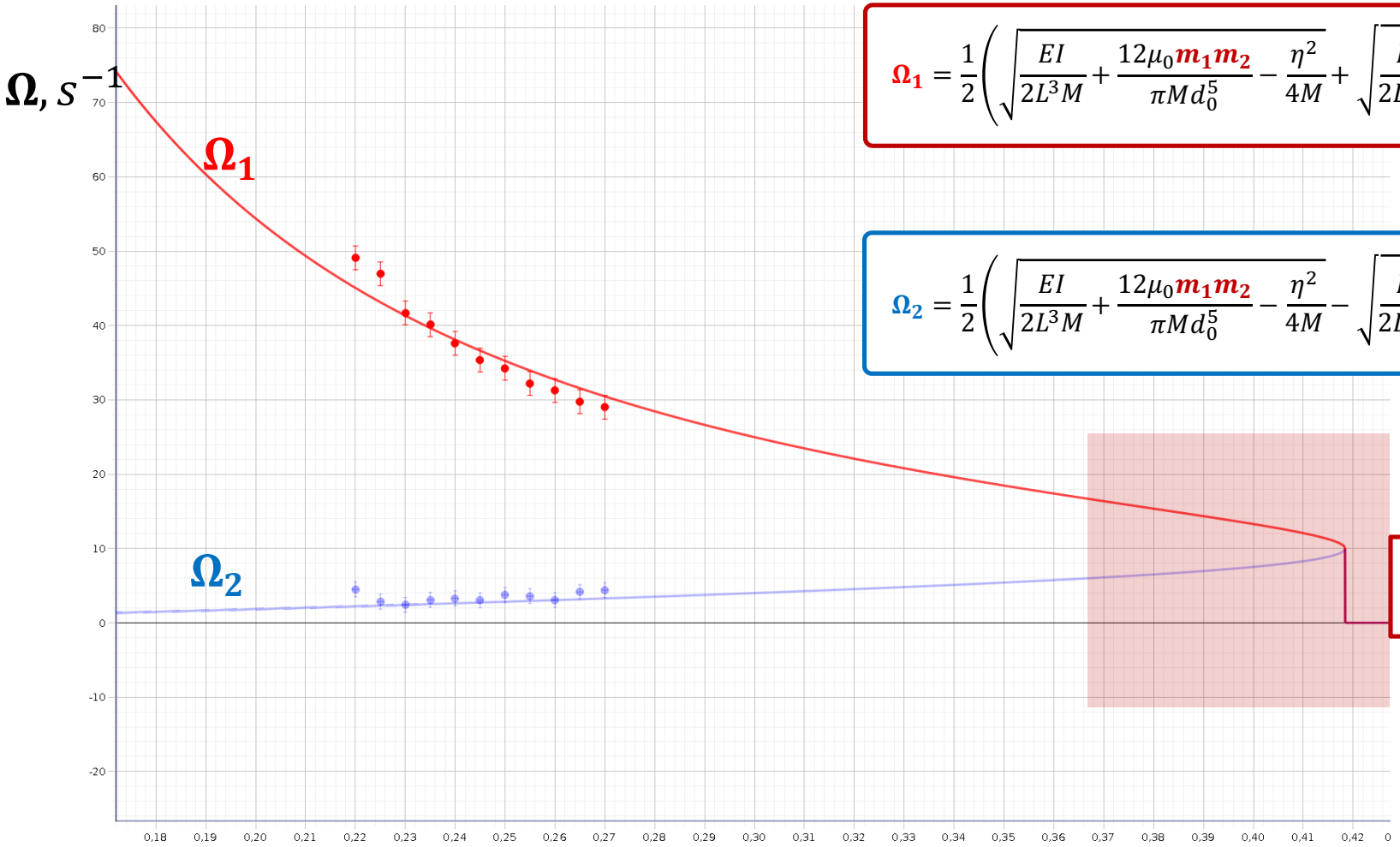




$$\Omega_1 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} + \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

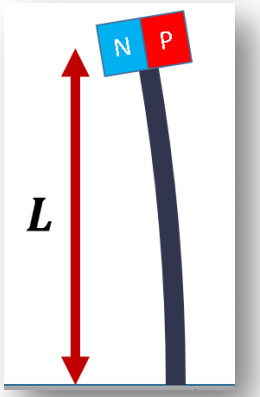
$$\Omega_2 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} - \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

Ω_1, Ω_2 *vs Length of leaf springs*



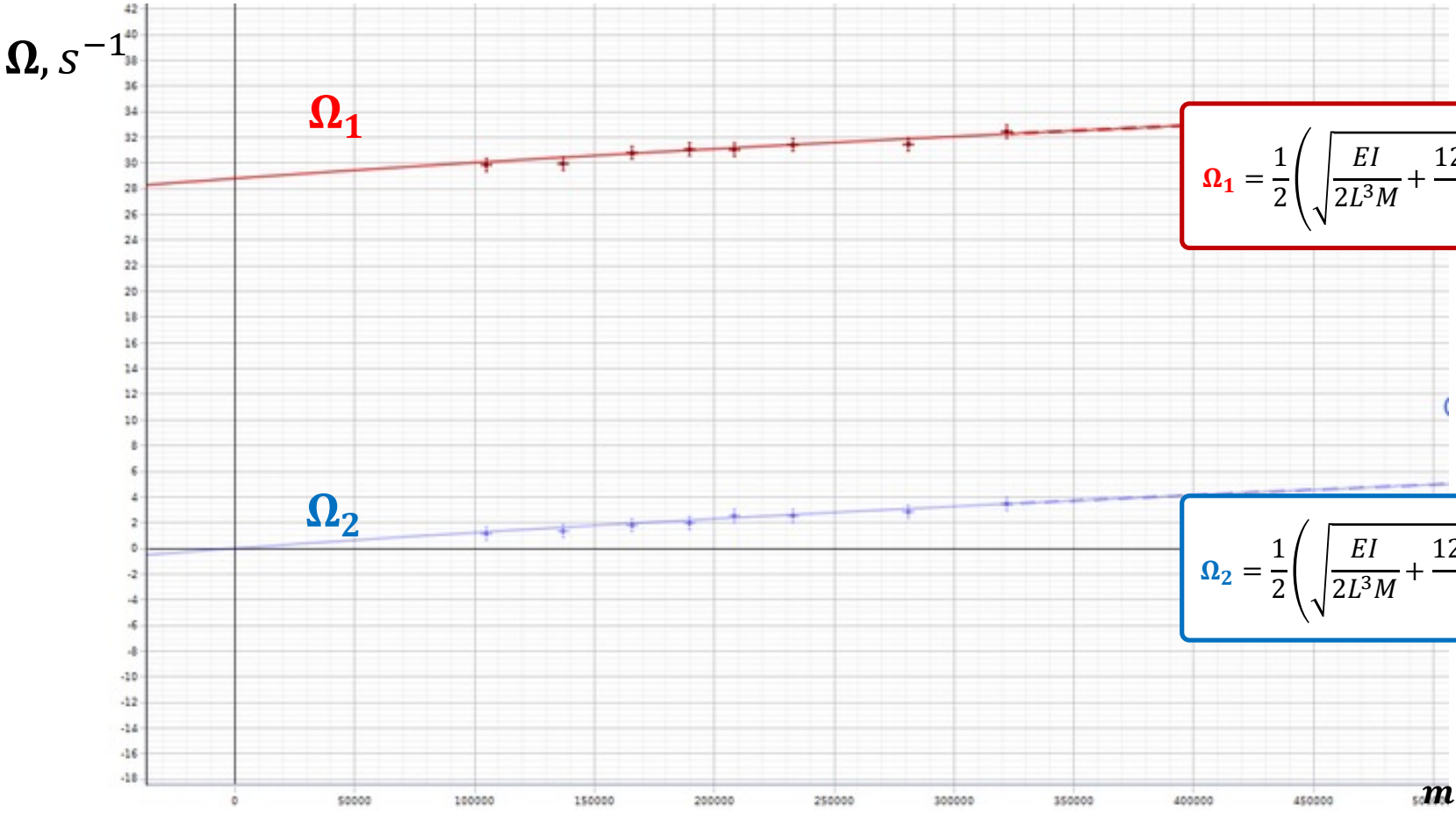
$$\Omega_1 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} + \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

$$\Omega_2 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} - \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$



Oscillations are damping before one period

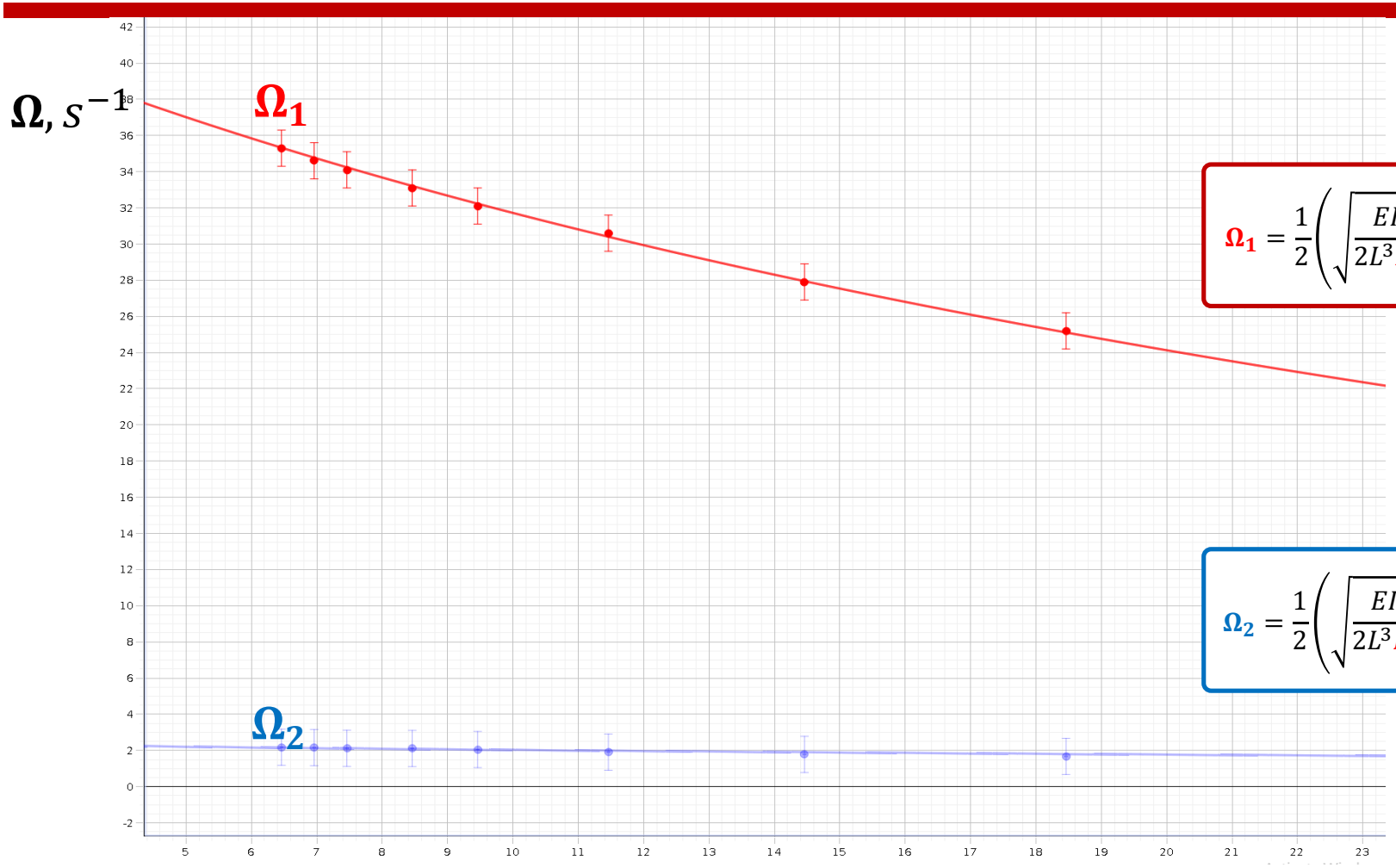
Ω_1, Ω_2 *vs multiple of magnetic moments*



$$\Omega_1 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} + \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

$$\Omega_2 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} - \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

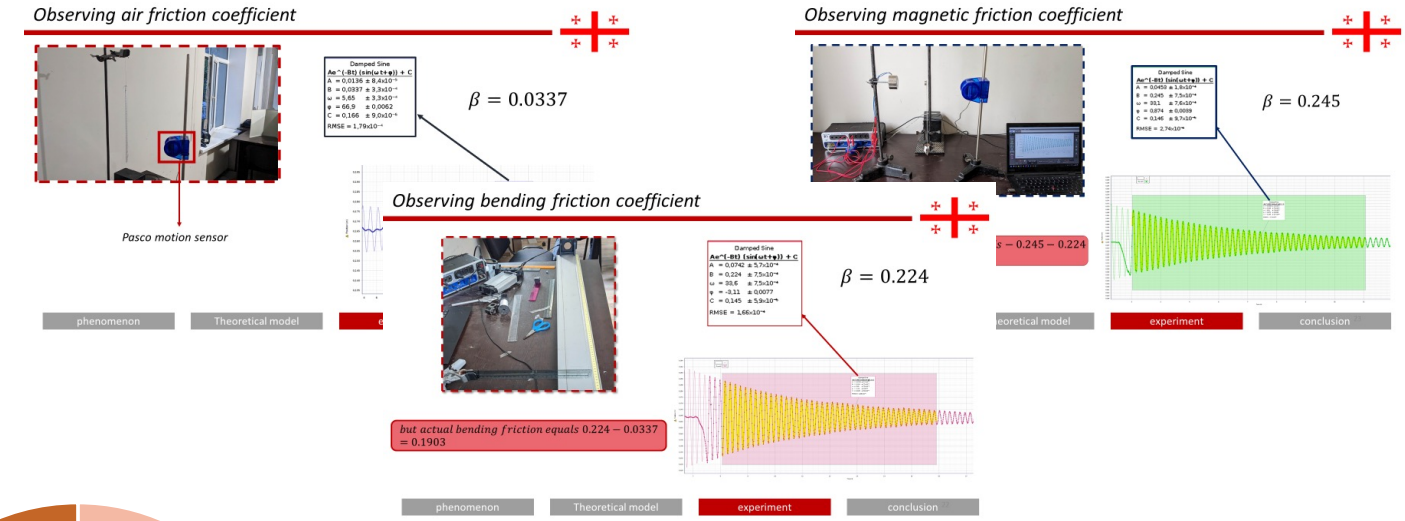
Ω_1, Ω_2 *vs mass of the magnets*

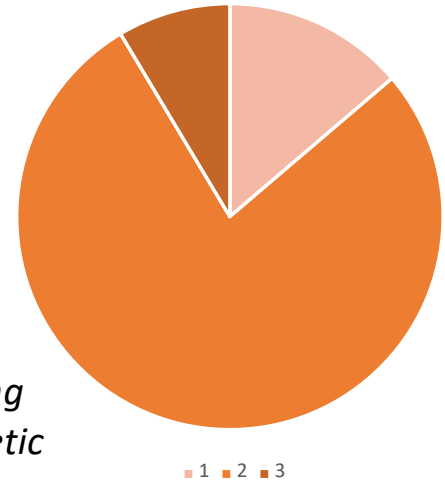


$$\Omega_1 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} + \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

$$\Omega_2 = \frac{1}{2}\left(\sqrt{\frac{EI}{2L^3M} + \frac{12\mu_0 m_1 m_2}{\pi M d_0^5} - \frac{\eta^2}{4M}} - \sqrt{\frac{EI}{2L^3M} - \frac{\eta^2}{4M}}\right)$$

Comparing all friction coefficients

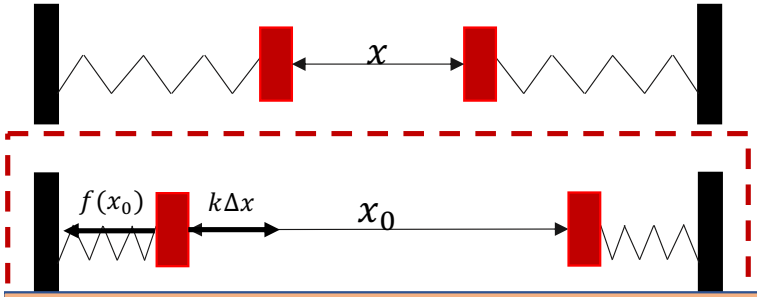




1.air
2.Bending
3.magnetic

The main friction force acting on the system is springs bending coefficient

Characteristic of force between magnets



$$m\ddot{x}_1 = k(x_0 - x - x_1) - f(x_0 - x_1 + x_2)$$
$$m\ddot{x}_2 = -k(x_0 - x + x_2) + f(x_0 - x_1 + x_2)$$

→ *Initial motion equations*

$$f(x_0 - (x_1 - x_2)) \approx f(x_0) + (x_1 - x_2)f'(x_0)$$

$$f = f(x_0) + \Delta x f'(x_0) + \alpha f \Delta x^2 + \beta f'''(x)\Delta x^3$$

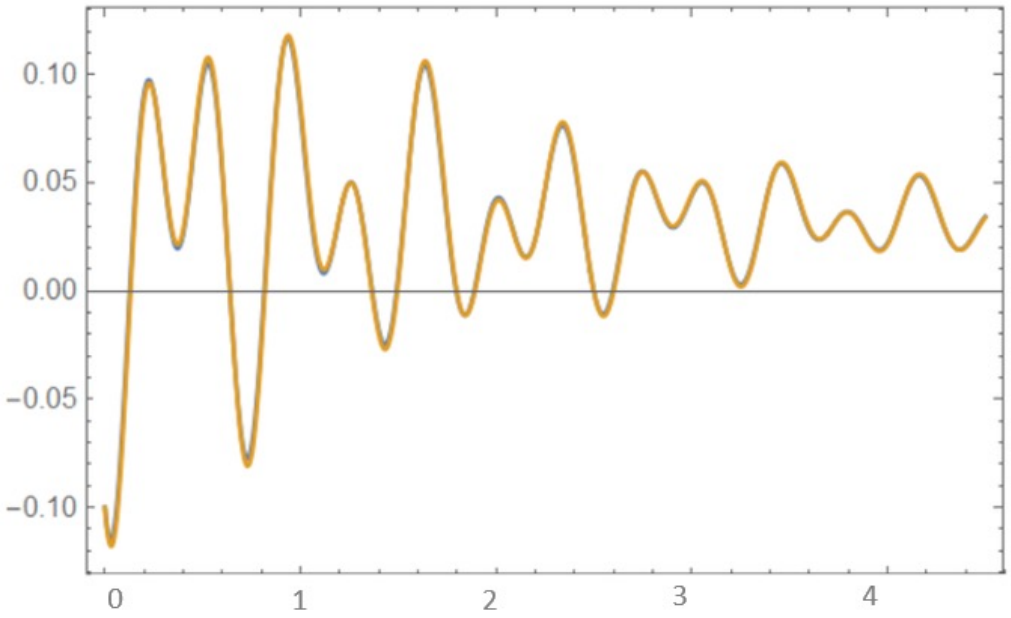
$$m\ddot{x}_1 = -kx_1 - (x_1 - x_2)f'(x_0) - f(x_0) + k\Delta x$$
$$m\ddot{x}_2 = -kx_2 + (x_1 - x_2)f'(x_0) - f(x_0) + k\Delta x$$

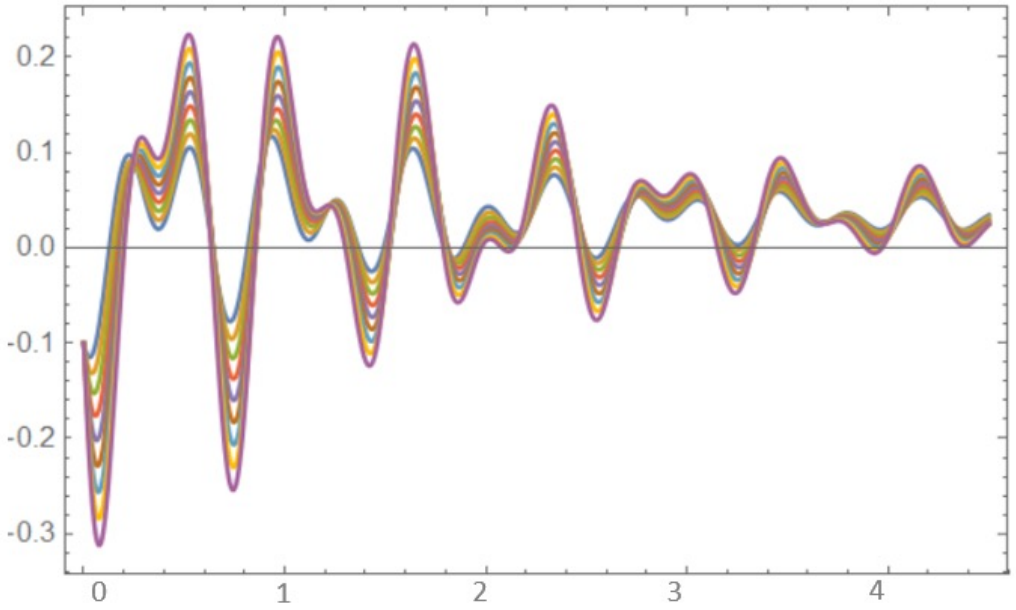
→ The total force acting on mass in stable situation, equals to zero

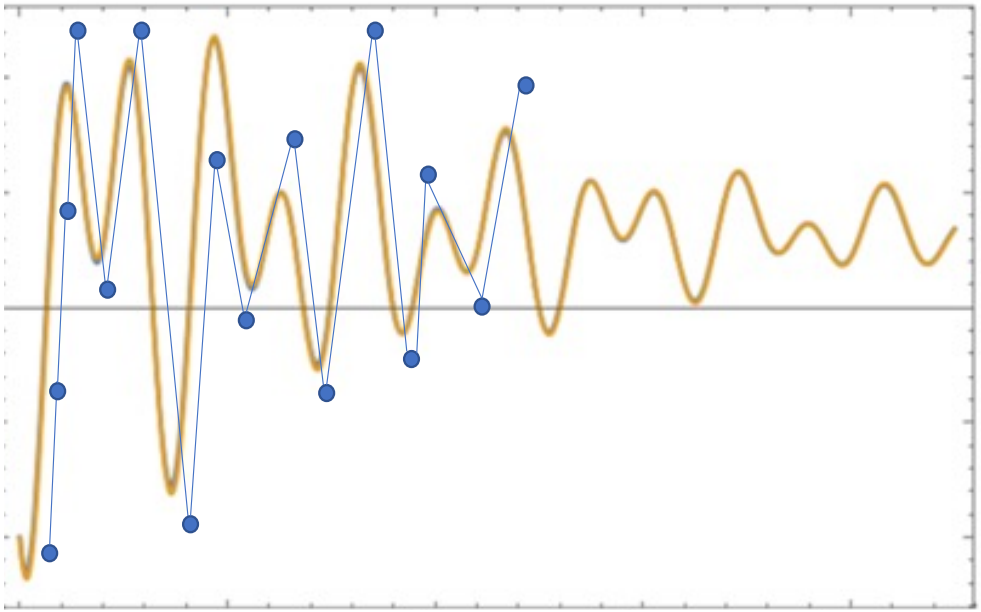
Final motion equations

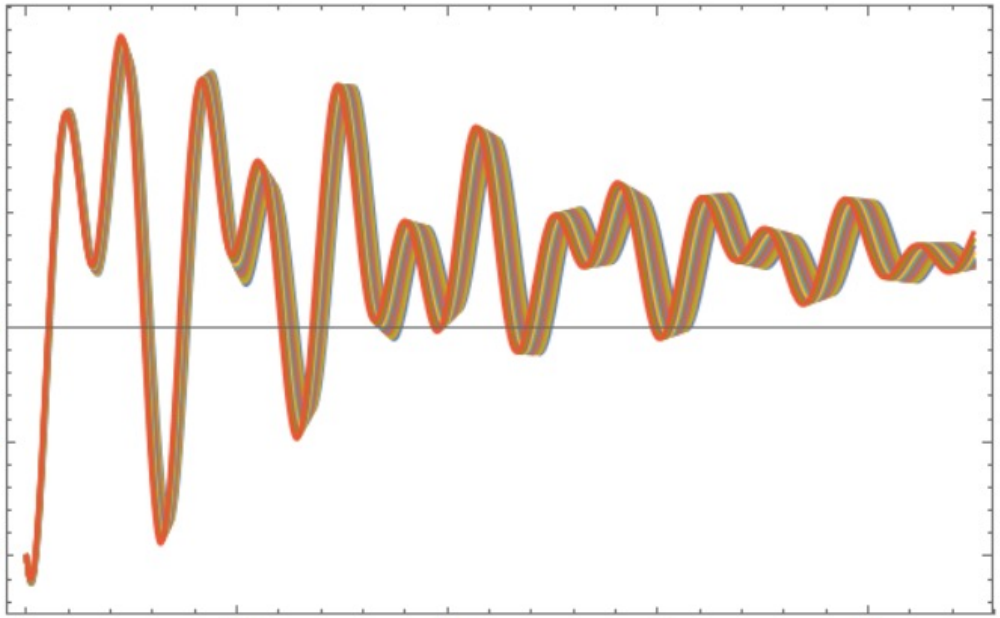
$$m\ddot{x}_1 = -kx_1 - (x_1 - x_2)f'(x_0)$$
$$m\ddot{x}_2 = -kx_2 + (x_1 - x_2)f'(x_0)$$

Not sensitive to initial conditions



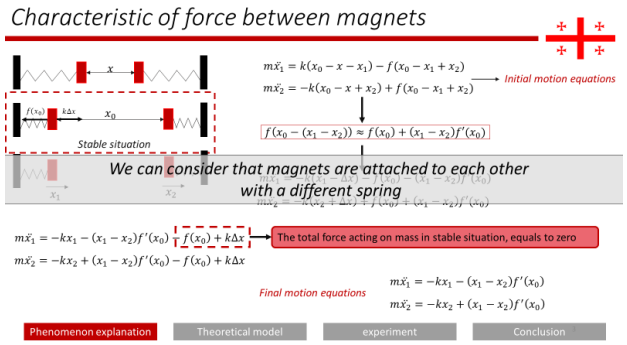






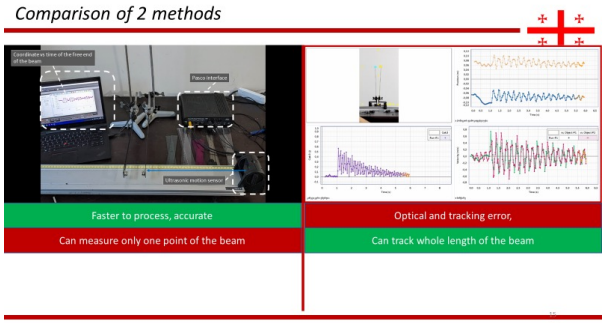
Conclusion

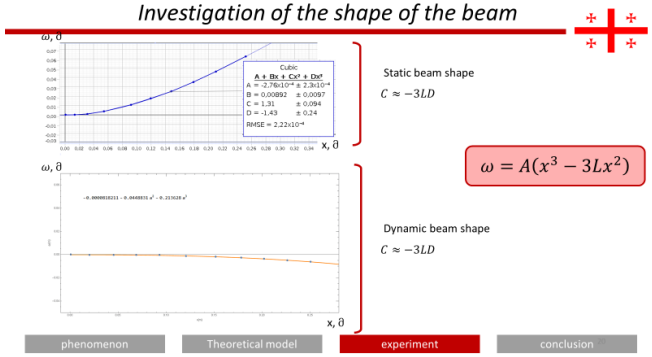
Conclusion



We explained why can we consider that the magnets are attached with spring to each other

Presented 2 methods of data tracking and compared them

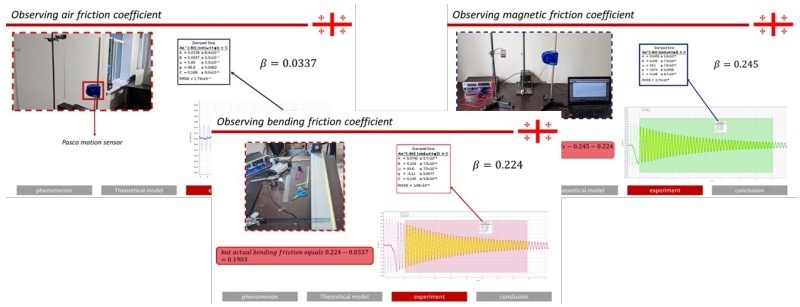




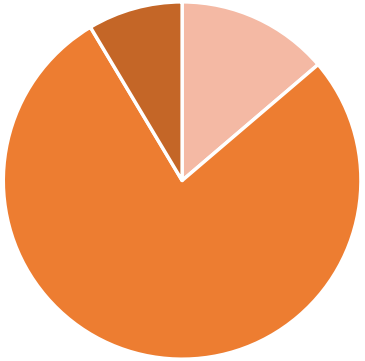
We proved that we could use the static beam theory for our system springs with a good approximation

phenomenon | Theoretical model | experiment | conclusion

Conclusion

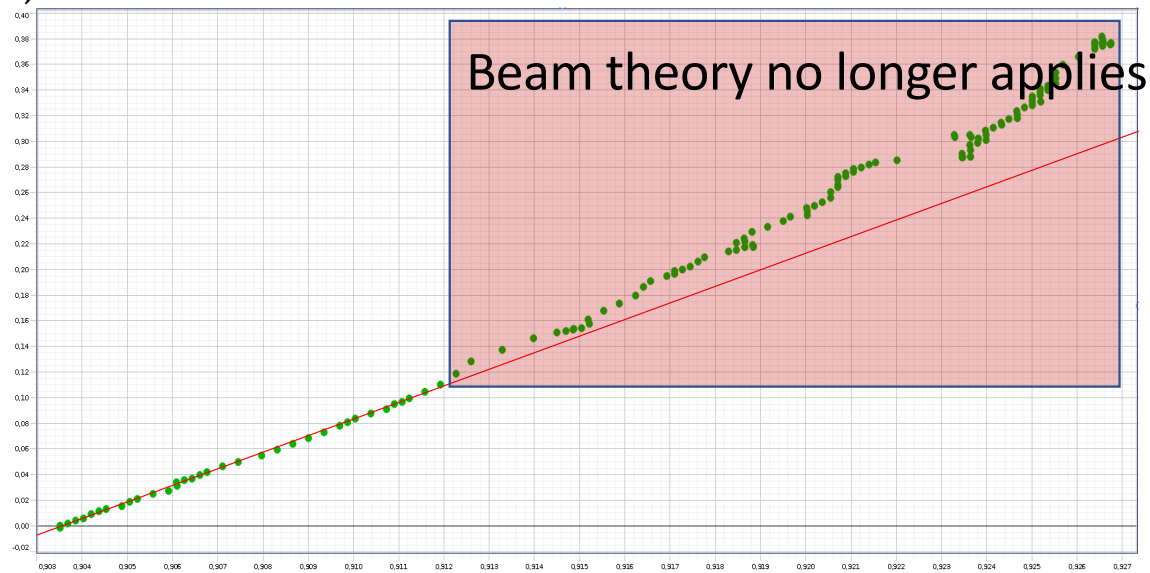


1.air
2.Bending
3.magnetic



The main friction force acting on the system is springs bending coefficient

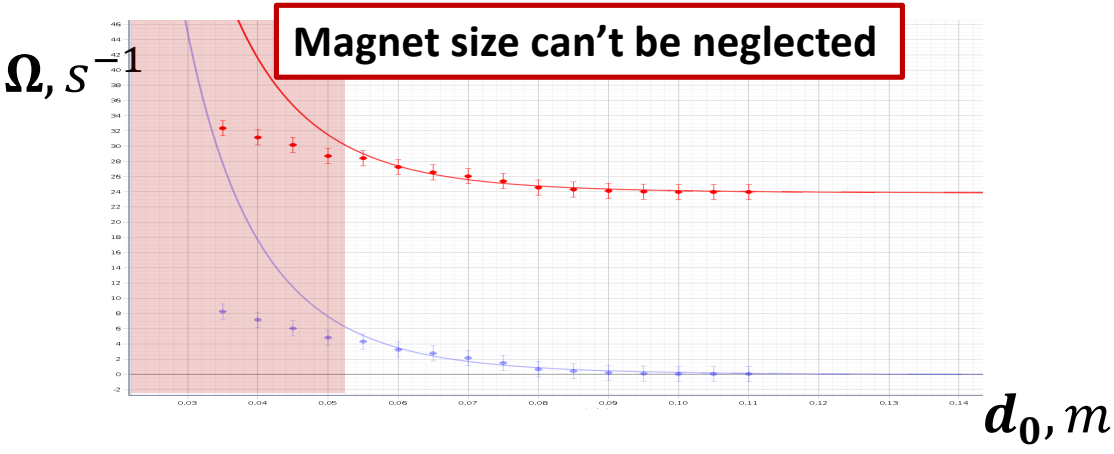
F, n



Measured the critical displacement when our beam theory no longer applies

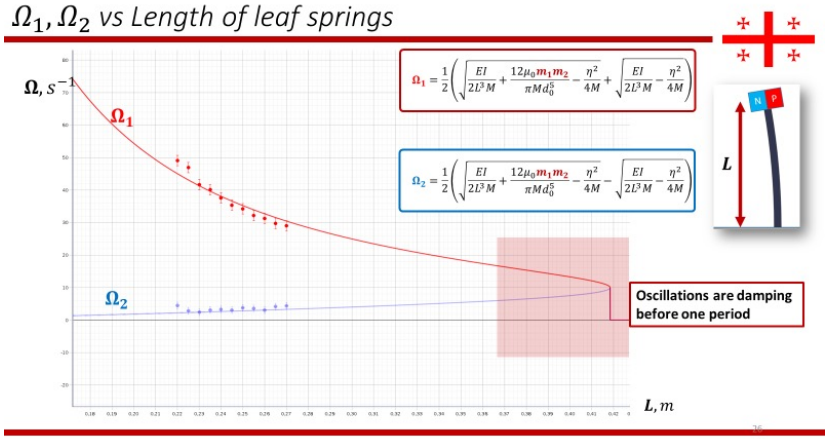
phenomenon | Theoretical model | experiment | conclusion

Conclusion



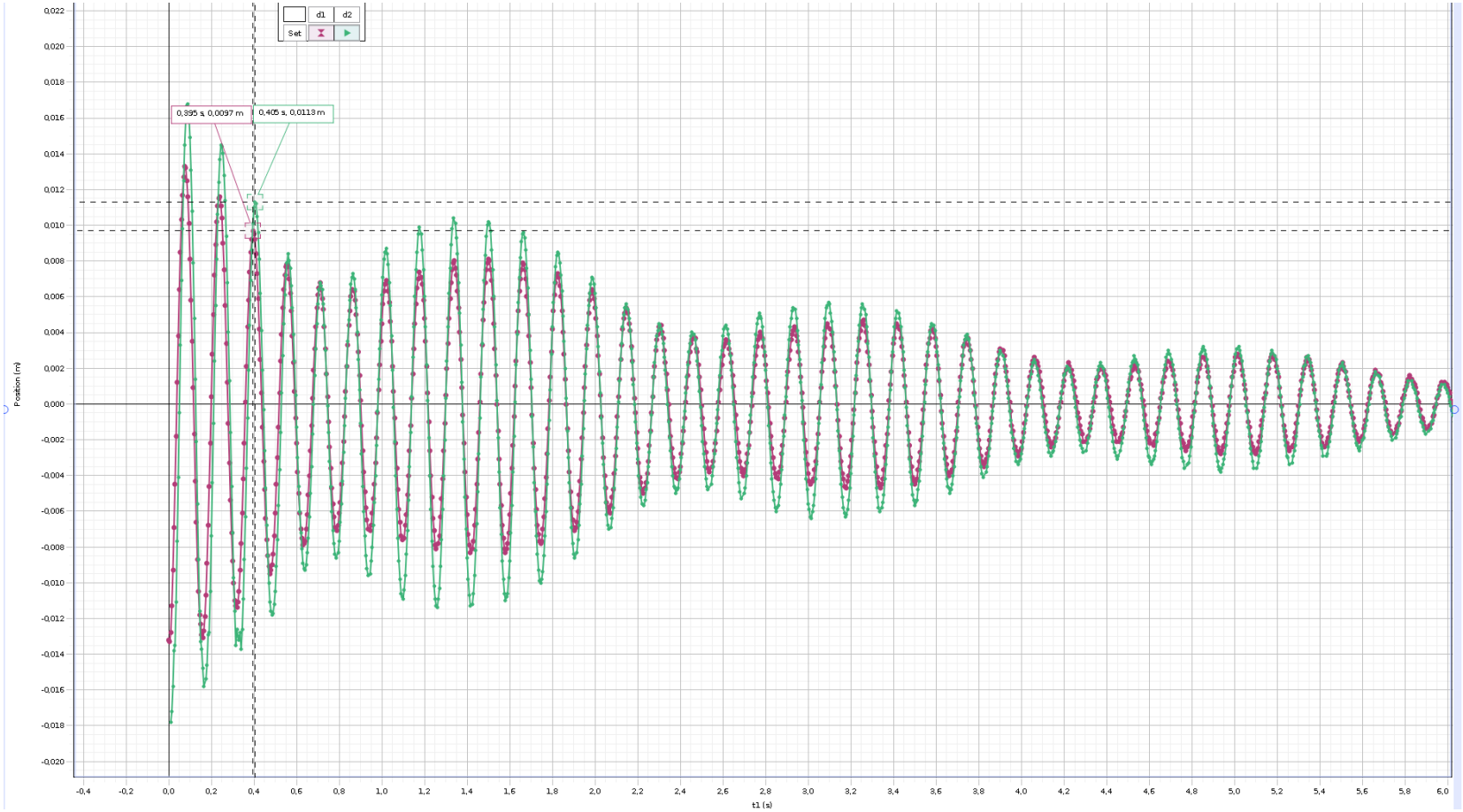
Found a critical distance between magnets for our theory

Found a critical length of springs for our theory



phenomenon | Theoretical model | experiment | conclusion

Thanks for attention!


d1
d2
Set
0,395 s, 0,0097 m
0,405 s, 0,0113 m
Position (m)
t1 (s)